Available online at www.sciencedirect.com

SCIENCE DIRECT®

Earth and Planetary Science Letters 237 (2005) 647–672

EPSL

www.elsevier.com/locate/epsl

Paleomagnetism and $^{40}Ar/^{39}Ar$ Geochronology of Yemeni Oligocene volcanics: Implications for timing and duration of Afro-Arabian traps and geometry of the Oligocene paleomagnetic field

Peter Riisager [a,b,c,*], Kim B. Knight [d], Joel A. Baker [c], Ingrid Ukstins Peate [c,e], Mohamed Al-Kadasi [f], Abdulkarim Al-Subbary [f], Paul R. Renne [d,g]

[a] *University of Lund, GeoBiosphere Science Centre, Sölvegatan 12, S-223 62 Lund, Sweden*
[b] *University of Copenhagen, Geological Museum, Øster Voldgade 5-7, DK-1350 Copenhagen K, Denmark*
[c] *Danish Lithosphere Centre, Øster Voldgade 10, DK-1350 Copenhagen K, Denmark*
[d] *University of California, Berkeley, Department of Earth and Planetary Science, Berkeley, CA 94720, USA*
[e] *University of Iowa, Geoscience Department, 121 Trowbridge Hall, Iowa City, IA 52242, USA*
[f] *Sana'a University, Department of Geology, P.O. Box 1247, Sana'a, Yemen*
[g] *Berkeley Geochronology Center, 2455 Ridge Road, Berkeley, CA 94709, USA*

Received 11 February 2005; received in revised form 16 April 2005; accepted 2 June 2005
Available online 2 August 2005
Editor: V. Courtillot

Abstract

A combined paleomagnetic and $^{40}Ar/^{39}Ar$ study was carried out along eight stratigraphically overlapping sections in the Oligocene Afro-Arabian flood volcanic province in Yemen (73 sites). The composite section covers the entire volcanic stratigraphy in the sampling region and represents five polarity zones that are correlated to the geomagnetic polarity time scale based on $^{40}Ar/^{39}Ar$ ages from this and previous studies. The resulting magnetostratigraphy is similar to that of the conjugate margin in Ethiopia. The earliest basaltic volcanism took place in a reverse polarity chron that appears to correspond to C11r, while the massive rhyolitic ignimbrite eruptions correlated to ash layers in Oligocene Indian Ocean sediment 2700 km away from the Afro-Arabian traps, appear to have taken place during magnetochron C11n. The youngest ignimbrite was emplaced during magnetochron C9n. Both $^{40}Ar/^{39}Ar$ and paleomagnetic data suggest rapid <1 Ma eruption of the basal basalt units and punctuated eruption of the upper silicic units over a duration potentially as long as 3 Ma with interspersed eruptive hiatuses. Eruption of the basal basalts may have preceded the Oi2 cooling event.

The paleomagnetic pole $\lambda = 74.2°$N, $\varphi = 249.1°$E (A95 $= 3.6°$; $N = 48$) is supported by a positive reversal test. Paleosecular variation, estimated as the angular standard deviation of the VGP distribution $14.2° + 2.3° / - 1.7°$, is close to expected, suggesting that the paleomagnetic pole represents a time-averaged field. The pole is in excellent accord with the

* Corresponding author. University of Lund, GeoBiosphere Science Centre, Sölvegatan 12, S-223 62 Lund, Sweden.
E-mail address: peter.riisager@geol.lu.se (P. Riisager).

0012-821X/$ - see front matter © 2005 Elsevier B.V. All rights reserved.
doi:10.1016/j.epsl.2005.06.016

paleomagnetic poles obtained from the Ethiopian part of the Afro-Arabian province, after closure of the Red Sea. By analyzing Afro-Arabian paleomagnetic data in conjunction with contemporaneous paleomagnetic poles available from different latitudes we argue that the Oligocene paleomagnetic field was dominated by the axial dipole with insignificant non-dipole field contributions.
© 2005 Elsevier B.V. All rights reserved.

Keywords: Yemen; Afro-Arabian traps; Oligocene; $^{40}Ar/^{39}Ar$ geochronology; magnetostratigraphy; paleomagnetic pole; large igneous province

1. Introduction

Large igneous provinces (LIPs) represent major geodynamic events associated with regional-scale uplift, massive volcanism, and in some cases continental rifting and breakup [1]. Geochronology and magnetostratigraphy studies have played an important role in our understanding of LIPs, particularly in documenting the very rapid emplacement in only a few Ma for most LIPs (e.g., [1,2]), indicating that LIP occurrences may be related to different and more dynamic mantle convection processes than those driving the Earth's normal-state plate tectonics. Geochronology and magnetostratigraphy data have also documented synchrony between many LIP occurrences and major global environmental changes including biotic mass extinctions, suggesting that LIP eruptions may be capable of causing global environmental and climatic effects [2,3].

This study presents new paleomagnetic, magnetostratigraphic and $^{40}Ar/^{39}Ar$ geochronology data from the Oligocene Afro-Arabian LIP in Yemen. The mag-

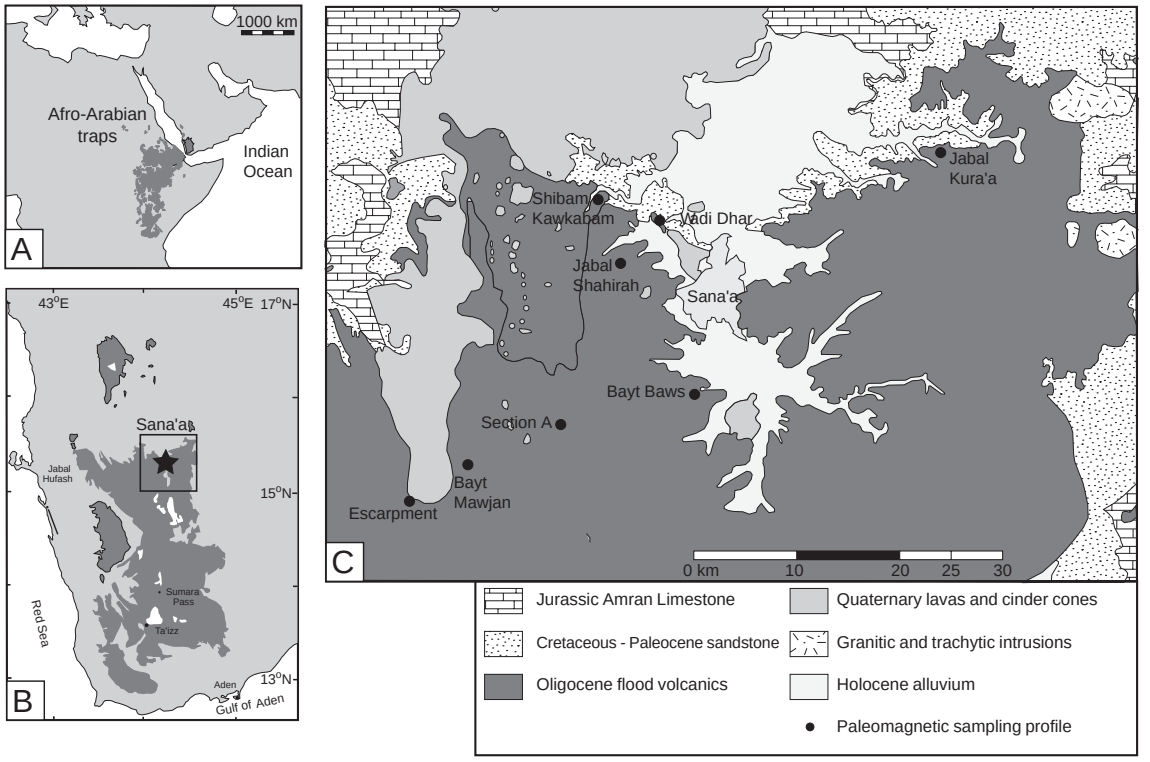


Fig. 1. Distribution of the Oligocene flood volcanism (in dark grey) on maps of (A) Afro-Arabia traps, and (B) Yemen with star indicating the location of Sana'a. (C) Detailed geological map of the sampling region around the Yemeni capital Sana'a. The exact geographic coordinates of the sampling profiles (black dots) are given in Table 1.

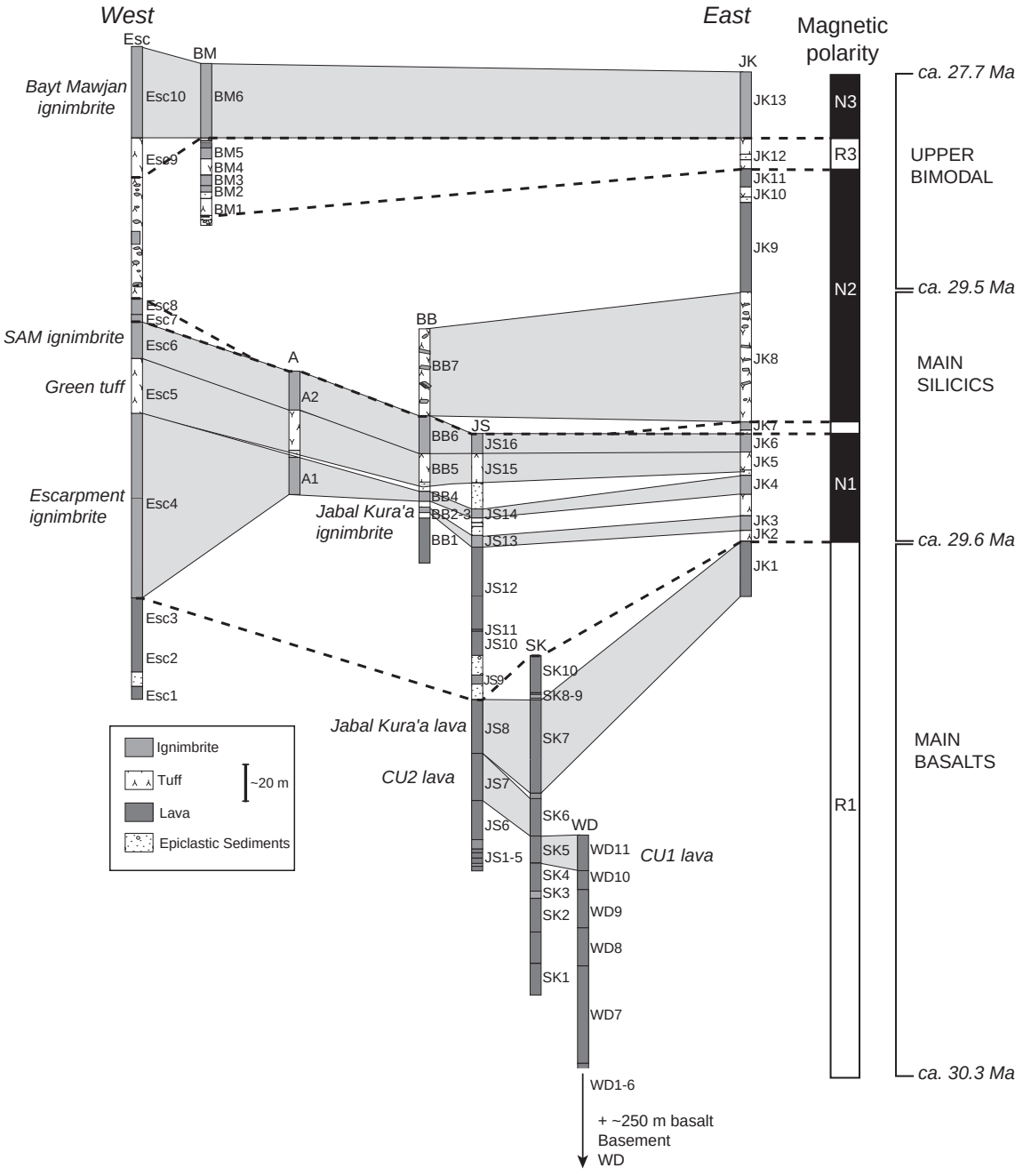


Fig. 2. Paleomagnetic sampling profiles (see Fig. 1) and their volcanostratigraphic correlations. Section abbreviations, from west to east: JK — Jabal Kura'a, WD — Wadi Dhar, SK — Shibam Kawkabam, JS — Jabal Shahirah, BB — Bayt Baws, A — Section A, BM — Bayt Mawjan, Esc — Escarpment. The Wadi Dhar section continues with a further 250 m of basaltic lavas beneath those shown. The magnetic polarity zonations and volcanostratigrapic divisions are given to the right.

Table 1
Site name and lithology (stratigraphy in Fig. 2); Lat., Long. and Alt. are the geographic latitudes and longitudes and altimeter altitude readings; Strike/dip is the structural attitudes based on in-field estimates using a Brunton compass; N/n is the number of treated/used samples; Age $\pm 2\sigma$ is the determined $^{40}Ar/^{39}Ar$ age and associated uncertainty

a

Site	Lithology	Lat (N)	Long (E)	Alt (m)	Strike/dip	n/N	Age $\pm 2s$ (Ma)	I_{geo} (°)	D_{geo} (°)	k	a_{95} (°)	I_{str} (°)	D_{str} (°)
Escarpment													
Esc1	Lava	15.169	43.972	2050	0/3	6/6		11.9	181.9	87.2	7.2	12.0	181.3
Esc2	Lava	15.169	43.940	2065	0/3	6/6		15.3	174.7	228.8	4.4	15.0	173.9
Esc3	Lava	15.168	43.938	2098	0/3	6/6		19.3	151.7	59.2	8.8	17.9	150.8
Esc4*	Ignimbrite	15.178	43.936	2200	160/7	17/17		−2.7	348.0	164.3	2.8	4.2	348.0
Esc5*	Tuff	15.161	43.945	2435	160/7	7/9		−12.6	6.9	510.1	2.7	−6.3	6.4
Esc6*	Ignimbrite	15.161	43.945	2450	160/7	8/8	29.47 ± 0.14 (2)	−7.9	4.4	223.1	3.7	−1.5	4.2
Esc7	Ignimbrite	15.160	43.945	2470	160/7	3/3		18.9	164.4	231.3	8.1	11.9	164.3
Esc8	Ignimbrite	15.161	43.945	2480	160/7	6/7		−23.8	171.7	381.4	3.4	−30.6	172.5
Esc9	Tuff	15.159	43.943	2530	160/7	5/6		21.5	351.0	88.1	8.2	28.4	351.6
Esc10*	Ignimbrite	15.154	43.948	2655	160/7	6/6		12.9	27.9	476.7	3.1	18.1	26.6
Bayt Mawjan													
BM1	Tuff	15.201	43.983	2800	40/20	4/4	28.40 ± 0.14 (3)	24.1	174.4	200.5	6.5	9.2	170.3
BM2	Ignimbrite	15.202	43.981	2810	40/20	7/7		50.0	169.5	310.2	3.4	33.4	159.3
BM3	Ignimbrite	15.202	43.980	2835	40/20	7/7		42.7	159.4	462.7	2.8	24.7	153.4
BM4	Tuff	15.200	43.980	2811	40/20	3/9		14.8	176.5	249.8	7.8	0.7	174.5
BM5	Ignimbrite	15.203	43.980	2840	40/20	6/6	28.00 ± 0.10 (3)	−18.1	166.7	545.4	2.9	−33.5	173.0
BM6*	Ignimbrite	15.202	43.979	2860	40/20	5/6	27.67 ± 0.12 (1)	−5.8	350.8	38.4	12.5	9.4	351.2
Section A													
A1*	Ignimbrite	15.286	44.074	2825	250/6	8/8		6.2	3.4	142.8	4.7	0.7	3.3
A2*	Ignimbrite	15.283	44.076	2905	250/6	7/7		1.7	0.9	263.0	3.7	−3.9	0.9
Bayt Baws													
BB1	Lava	15.271	44.199	2400	0/0	0/0	29.61 ± 0.08 (1)						
BB2	Tuff	15.271	44.199	2410	0/0	0/0							
BB3*	Ignimbrite	15.271	44.199	2425	0/0	4/7		−7.7	6.8	179.5	6.9	−7.7	6.8
BB4*	Ignimbrite	15.274	44.199	2440	0/0	14/14		−9.8	345.8	85.0	4.3	−9.8	345.8
BB5*	Tuff	15.272	44.200	2440	0/0	4/4	29.59 ± 0.12 (1)	19.0	355.5	53.3	12.7	19.0	355.5
BB6*	Ignimbrite	15.272	44.200	2452	0/0	38/45		−0.8	3.7	83.0	2.6	−0.8	3.7
BB7	Tuff	15.272	44.200	2470	0/0	0/0	29.48 ± 0.13 (1)						
Jabal Shahirah													
JS1	Lava	15.389	44.138	2435	30/3	4/4		25.7	149.8	537.6	4.0	23.1	149.1
JS2	Lava	15.386	44.135	2424	30/3	6/6		16.8	180.0	344.6	3.6	15.3	179.3
JS3	Lava	15.387	44.135	2432	30/3	7/8		6.0	188.5	82.4	6.7	4.9	188.2
JS4	Lava	15.387	44.136	2445	30/3	8/8		−1.4	181.0	245.5	3.5	−2.9	181.1
JS5	Tuff	15.385	44.135	2440	30/3	0/3							
JS6	Lava	15.385	44.135	2455	30/3	7/7	29.94 ± 0.24 (1)	−1.1	176.5	29.8	11.2	−2.8	176.6
JS7*	Lava	15.383	44.136	2475	30/3	15/16		−9.3	175.2	192.4	2.8	−11.0	175.6
JS8*	Lava	15.384	44.135	2510	30/3	7/7		−7.3	176.4	367.5	3.2	−9.0	176.8
JS9	Tuff	15.382	44.138	2550	30/3	2/6		19.1	0.9			20.5	1.8
JS10	Silicic lava	15.381	44.137	2580	30/3	5/7		−21.9	350.5	170.4	5.9	−20.0	349.6
JS11	Tuff	15.380	44.136	2590	30/3	6/8		39.0	350.9	81.4	7.5	40.9	352.9
JS12	Lava	15.379	44.136	2640	30/3	3/8		11.5	345.6	78.4	14.0	13.6	346.1
JS13*	Ignimbrite	15.379	44.137	2660	30/3	6/8		−3.6	6.5	93.8	7.0	−2.4	6.4
JS14*	Ignimbrite	15.379	44.137	2675	30/3	3/3		9.1	352.4	193.1	8.9	10.9	352.8

Table 1 (*continued*)

a

Site	Lithology	Lat (N)	Long (E)	Alt (m)	Strike/dip	*n*/*N*	Age ±2s (Ma)	I_{geo} (°)	D_{geo} (°)	*k*	a_{95} (°)	I_{str} (°)	D_{str} (°)
Jabal Shahirah													
JS15*	Tuff	15.378	44.136	2708	30/3	9/9		−0.4	1.3	367.4	2.7	1.0	1.3
JS16*	Ignimbrite	15.377	44.136	2725	30/3	8/11		−2.2	357.2	95.3	5.7	−0.6	357.1
Shibam Kawkabam													
SK1†	Lava	15.417	44.091	2550	20/4	8/8	30.24 ± 0.18 (1)	−13.2	119.8	100.2	5.6	−17.1	120.0
SK2	Lava	15.413	44.110	2465	20/4	6/6		−18.1	184.6	344.3	3.6	−19.1	185.9
SK3	Ignimbrite	15.416	44.093	2551	20/4	0/0	30.16 ± 0.13 (2)						
SK4†	Lava	15.416	44.093	2550	20/4	4/4	30.20 ± 0.26 (1)	66.1	169.4	247.7	5.8	63.8	162.3
SK5*	Lava	15.416	44.093	2550	20/4	7/7		12.2	179.4	664.5	2.3	10.8	178.6
SK6*	Lava	15.417	44.094	2550	20/4	6/6	30.51 ± 0.36 (1)	−9.7	171.4	128.8	5.9	−11.6	172.1
SK7*	Lava	15.417	44.095	2550	20/4	5/9	30.03 ± 0.26 (1)	−6.2	166.3	28.9	14.5	−8.4	166.7
SK8	Dyke	15.415	44.098	2631	20/4	8/8		15.6	355.2	130.3	4.9	17.2	355.2
SK9	Tuff	15.415	44.098	2635	20/4	4/4		11.4	156.9	48.8	13.3	8.7	156.4
SK10	Lava	15.415	44.098	2640	20/4	7/8	30.30 ± 0.14 (1)	−8.7	169.7	31.3	10.9	−10.7	170.3
Wadi Dhar													
WD1	Lava	15.432	44.121	2330	15/3	5/5		−26.6	163.9	160.5	6.1	−28.1	165.2
WD2	Lava	15.431	44.121	2370	15/3	9/11		19.4	176.2	40.1	8.2	18.4	175.2
WD3	Lava	15.430	44.122	2370	15/3	5/9		−29.4	173.6	84.0	8.4	−30.5	175.2
WD4	Lava	15.431	44.121	2405	15/3	6/7		−18.9	171.6	465.4	3.1	−20.1	172.6
WD5	Lava	15.408	44.117	2420	15/3	6/6		−3.9	172.6	501.9	3.0	−5.0	172.8
WD6	Lava	15.409	44.116	2415	15/3	12/12		3.9	170.2	175.6	3.3	2.6	170.0
WD7	Lava	15.409	44.116	2475	15/3	6/6		−5.0	168.0	596.4	2.7	−6.4	168.3
WD8	Lava	15.411	44.111	2420	15/3	8/8		6.6	183.9	266.7	3.4	6.0	183.6
WD9	Lava	15.412	44.112	2430	15/3	6/6		−26.9	180.5	342.1	3.6	−27.6	182.0
WD10	Lava	15.412	44.110	2440	15/3	7/7		−21.6	180.3	1331.3	1.7	−22.3	181.5
WD11*	Lava	15.413	44.110	2455	15/3	5/7		15.7	176.3	342.9	4.1	14.7	175.5
Jabal Kura'a													
JK1*	Lava	15.488	44.418	2435	0/0	6/7		−4.1	168.5	303.8	3.9	−4.1	168.5
JK2	Tuff	15.488	44.418	2460	0/0	7/7		−10.9	1.2	139.2	5.1	−10.9	1.2
JK3*	Ignimbrite	15.487	44.418	2470	0/0	7/7		−11.5	358.6	525.7	2.6	−11.5	358.6
JK4*	Ignimbrite	15.488	44.418	2480	0/0	11/11		−3.8	346.3	98.1	4.6	−3.8	346.3
JK5*	Tuff	15.488	44.418	2510	0/0	5/6		2.3	18.6	103.4	7.6	2.3	18.6
JK6*	Ignimbrite	15.488	44.418	2520	0/0	13/16		0.6	354.7	30.0	7.7	0.6	354.7
JK7	Ignimbrite	15.483	44.420	2530	0/0	5/6		45.5	160.9	476.3	3.5	45.5	160.9
JK8	Chaotic tuff	15.483	44.419	2545	0/0	9/14		9.3	343.7	15.0	13.7	9.3	343.7
JK9	Lava	15.482	44.418	2580	0/0	9/9		−2.2	351.9	151.0	4.2	−2.2	351.9
JK10	Tuff	15.482	44.418	2608	0/0	5/6		9.3	357.7	64.7	9.6	9.3	357.7
JK11	Lava	15.482	44.418	2610	0/0	0/8							
JK12	Tuff	15.479	44.419	2640	0/0	12/13		37.1	187.4	26.5	8.6	37.1	187.4
JK13*	Ignimbrite	15.479	44.419	2655	0/0	9/9		5.9	7.5	8.5	18.8	5.9	7.5

b

Cooling unit	Sites	*n*/*N*	*k*	a_{95} (°)	I_{str} (°)	D_{str} (°)
CU1 lava	SK5+WD11	12/14	344.4	2.3	12.4	177.4
CU2 lava	JS7+SK6	21/22	167.4	2.5	−11.1	174.6

(continued on next page)

Table 1 (*continued*)

b

Cooling unit	Sites	*n*/*N*	*k*	a_{95} (°)	I_{str} (°)	D_{str} (°)
Jabal Kura'a lava	JS8+SK7+JK1	18/23	72.1	4.1	−7.2	171.2
Jabal Kura'a ignimbrite	BB3+JS13+JK3	17/22	99.9	3.6	−7.4	3.3
Escarpment ignimbrite	Esc4+A1+BB4+JS14+JK4	53/53	57.9	2.6	−3.2	349.7
Green tuff	Esc5+BB5+JS15+JK5	25/28	36.8	4.8	1.2	5.8
SAM ignimbrite	Esc6+A2+BB6+JS16+JK6	74/87	62.9	2.1	−1.3	1.3
Bayt Mawjan ignimbrite	Esc10+BM6+JK13	20/21	11.7	10.0	10.9	9.1

Ages [1] are from present study, [2] are from correlative units in Ethiopia [6], and [3] are from Baker et al. [13]; I_{geo}, D_{geo} are in situ inclination and declination of flow-mean ChRMs; *k*, α_{95} are the precision parameter and the 95% confidence cone of the Fisher statistics [60]; I_{str}, D_{str} are inclination and declination of the tectonically corrected flow-mean ChRMs. Flows marked with asterisk (*) have been sampled along several units and are joined in Table 1b. Flows marked with a cross (†) record excursional directions.

netostratigraphy and geochronology results complement existing data from the Ethiopian part of the province, allowing a further discussion on the timing and duration of Oligocene Afro-Arabian LIP volcanism and its possible environmental and climatic effects. This study also presents a large data set of paleomagnetic directional data that allow a well-defined Oligocene paleomagnetic pole for Arabia to be

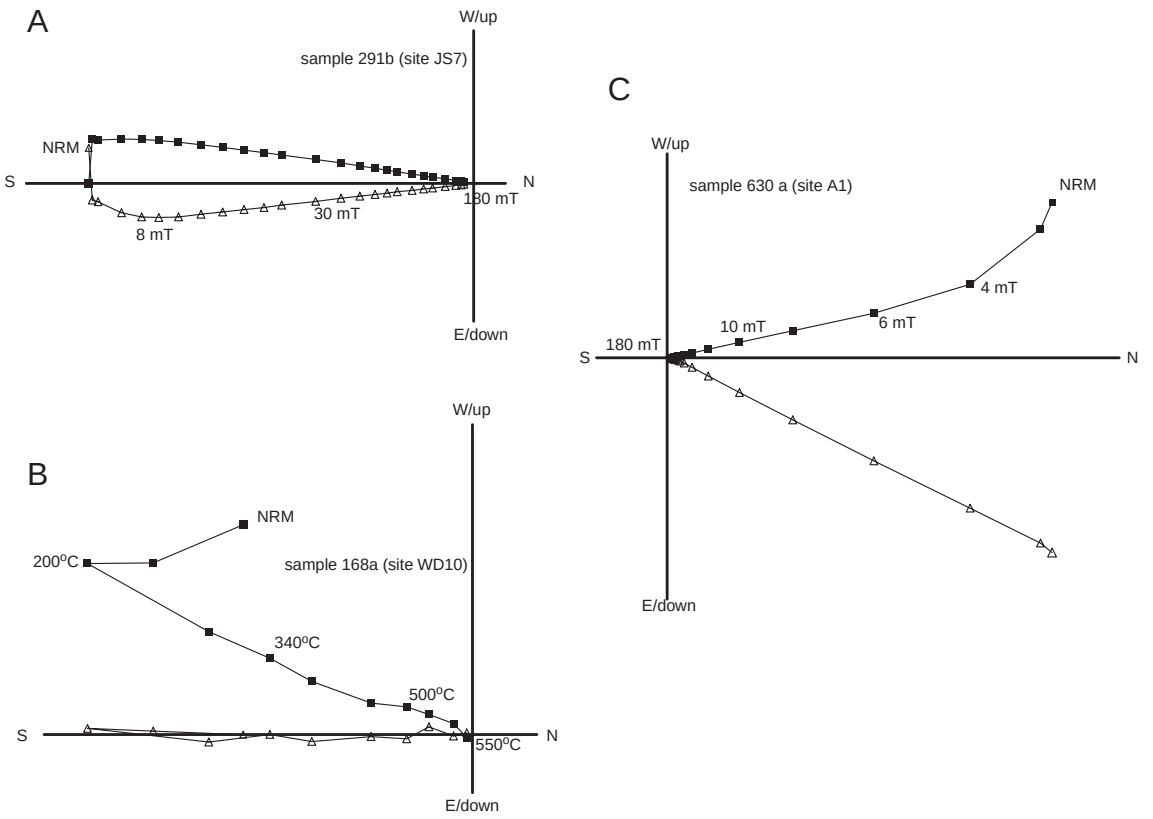


Fig. 3. Examples of orthogonal projections of alternating field and thermal demagnetizations. Solid (open) symbols on orthogonal projection correspond to vertical (horizontal) plane.

determined. Based on this pole we will discuss Afro-Arabian paleogeography and the Oligocene time-averaged field configuration.

2. Geology, stratigraphy and sampling

The Oligocene Afro-Arabian traps (Fig. 1A) represent one of the youngest and best preserved LIPs on Earth, with an original volume estimated to have been at least 1.2×10^6 km^2 [4]. Magnetostratigraphy and geochronology data from the Ethiopian part of the province have suggested that major basaltic flood volcanism in Ethiopia began about 31 Ma ago with the bulk of emplacement occurring over a period of 1–2 Ma [4–7]. This rapid emplacement is most easily explained by the plume-head or plume-impact model [8], in which widespread basaltic flood eruptions occur as the inflated head of a rising new mantle plume approaches the base of the lithosphere. A plume-impact origin is also supported by geochemical observations [9], the regional dome-shaped uplift between ~35 and ~30 Ma [10], and the anomalously low seismic shear velocity in the present-day Afro-Arabian upper mantle [11].

The Yemeni margin of the LIP (Fig. 1B) contains roughly 10% of the entire volume of erupted Oligocene Afro-Arabian flood volcanism [12]. Geochronology data suggest that volcanism in Yemen took place from 30.9 to 27.1 Ma ([13]; *this paper*). The earliest volcanic activity is composed entirely of basaltic lava flows, while more silicic volcanic products such as rhyolitic ignimbrites and tuffs dominate the middle part of the stratigraphy, followed by an upper bimodal sequence [14] (Fig. 2). Our sampling region around the capital city Sana'a (Fig. 1C) lies along the northern outskirts of the Afro-Arabian LIP where the volcanic sequence remains relatively untectonized and therefore more attractive for paleomagnetic studies compared to the extensively faulted and much thicker lava sequences in central parts of the province. The relatively simple geological setting also makes volcanostratigraphic correlation of sampling profiles possible through direct field observations. The eight stratigraphically overlapping sections that sampled for the present study cover the entire volcanic stratigraphy in the sampling region (Fig. 2), and their stratigraphic correlation has been confirmed by subsequent petrographic studies and geochemical correlations [14].

For the paleomagnetic study 587 oriented drill cores were collected from 71 sites (in average 8 cores per site). All paleomagnetic samples were drilled directly in the field with a portable gasoline

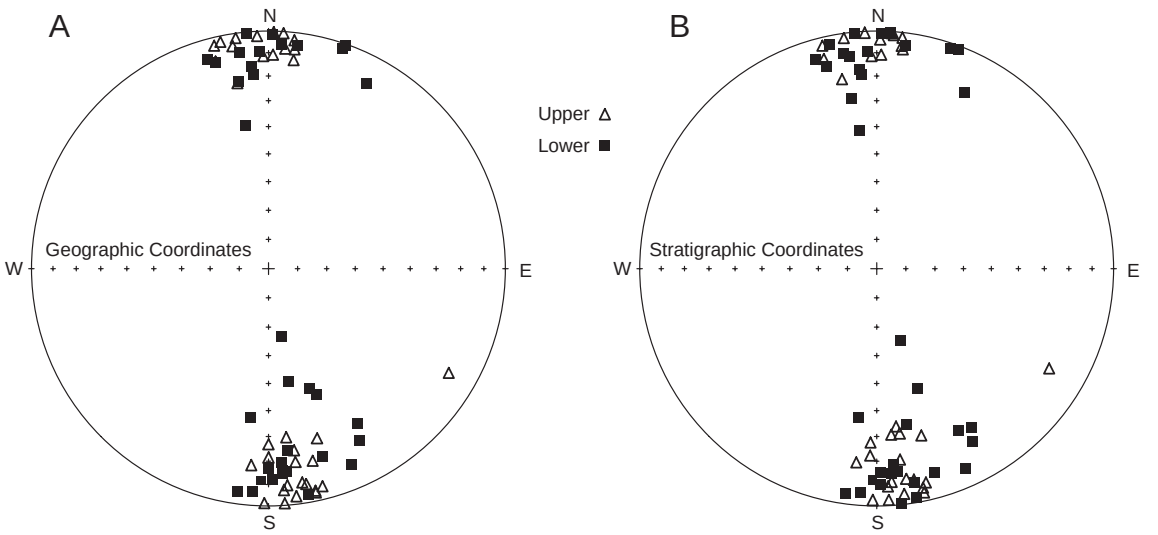


Fig. 4. Stereo diagram of paleomagnetic site–mean directions before (A) and after (B) tectonic correction. Open (solid) symbols correspond to negative (positive) inclination.

Table 2

[a]

Site	ID	Sample name	Disk: batch	Unit	Analyzed material	Size fraction (μm)	Plateau age (Ma)	±2σ	MSWD	^{39}Ar in plateau (%)	Total # of steps	Total gas age (Ma)	±2σ	Isochron age (Ma)	±2σ	^{40}Ar/^{36}Ar intercept	±2σ	MSWD
*	33274.1	J56	B:2	Rhyolite lava	K-feld	350–425	27.07	0.13	0.8	54.3	26	27.01	0.13	27.11	0.17	293	10	0.8
BM6	33277.1	YJB99.87	B:1	Bayt Mawjan ig.	K-feld	160–315	27.50	0.67	1.4	100.0	6	27.60	0.20	26.10	1.80	299	4	1.0
BM6	33283.1	YJB99.87	B:1	Bayt Mawjan ig.	K-feld	315–500	27.57	0.20	0.5	100.0	10	27.80	0.80	27.55	0.26	298	17	0.6
BM6	33277.2	YJB99.87	B:2	Bayt Mawjan ig.	K-feld	315–500	27.74	0.16	0.7	97.6	26	27.70	0.16	27.70	0.18	300	9	0.7
BB7	33257.1	YJB99.79	A:1	Iftar Alkalb tuff	K-feld	160–315	29.44	0.17	0.3	100.0	9	29.50	0.15	29.42	0.26	311	132	0.4
BB7	33282.1	YJB99.79	B:1	Iftar Alkalb tuff	K-feld	315–500	29.54	0.19	0.5	100.0	18	29.52	0.20	29.50	0.22	316	54	0.5
BB5	33271.1	YJB99.76	B:1	Green tuff	K-feld	315–500	29.56	0.17	0.5	54.9	10	29.48	0.16	29.58	0.18	247	229	0.6
BB5	33281.1	YJB99.76	B:1	Green tuff	K-feld	>500	29.62	0.17	0.6	100.0	10	29.61	0.17	29.65	0.19	264	97	0.6
BB2	33268.1	YJB99.70	B:3	Pumaceous tuff	K-feld	160–315	*No plateau*				20	29.61	0.14	29.58	0.14	*264*	*16*	4.7
BB2	33279.1	YJB99.70	B:4	Pumaceous tuff	K-feld	315–500	*No plateau*				23	29.89	0.14	29.79	0.14	*350*	*12*	48.0
ESC4	33261.1	YJB99.98	A:3	Escarpment ig.	Plag	160–315	*No plateau*				21	30.08	0.16	29.30	0.22	*307*	*2*	28.7
ESC4	33265.1	YJB99.98	A:4	Escarpment ig.	Plag	315–500	*29.73*	*0.10*	*1.0*	95.7	15	29.72	0.12	29.64	0.16	298	3	1.0
BB1	33263.1	YJB99.72	A:1	Akraban trachyan.	K-feld	160–315	29.59	0.12	1.1	62.7	9	29.55	0.10	29.49	0.16	316	57	2.4
BB1	33252.1	YJB99.72	A:1	Akraban trachyan.	K-feld	315–500	29.62	0.10	0.7	92.2	10	29.55	0.10	29.63	0.14	294	58	0.8
SK10	33308.1	YJB99.23	D:2	Basalt	Plag	160–315	30.21	0.26	0.6	100.0	25	30.10	0.30	29.84	0.48	315	20	0.5
SK10	33289.1	YJB99.23	C:4	Basalt	Plag	315–500	30.33	0.16	0.7	100.0	8	30.33	0.18	30.28	0.32	298	12	0.8
SK7	33316.1	YJB99.13	D:4	Kura'a basalt	Plag	160–315	30.03	0.26	0.9	100.0	11	30.10	0.30	29.98	0.32	300	16	1.0
SK7	33287.1	YJB99.13	C:2	Kura'a basalt	Plag	315–500	*29.76*	*0.18*	*0.4*	100.0	26	29.70	0.20	29.87	0.28	282	30	0.4
JS6	33302.1	YJB99.46	C:2	Basalt	Plag	315–500	29.94	0.24	0.7	100.0	26	29.90	0.30	29.76	0.46	300	8	0.6
SK6	33290.1	YJB99.26	C:4	Basalt	Plag	63–125	*No plateau*				7	30.70	0.40	29.75	1.10	*326*	*23*	7.0
SK6	33311.1	YJB99.26	D:2	Basalt	Plag	125–425	30.51	0.36	0.4	68.8	21	35.10	0.40	30.55	0.69	295	24	0.5
SK4	33314.1	YJB99.33	D:4	Basalt	Plag	160–315	30.21	0.28	0.8	100.0	7	30.20	0.30	29.99	0.38	303	8	0.4
SK4	33321.1	YJB99.33	D:2	Basalt	Plag	315–500	30.14	0.66	0.6	55.0	24	29.80	0.50	29.95	0.80	303	14	0.5
SK1	33312.1	YJB99.30	D:4	Basalt	Plag	160–315	30.37	0.24	0.8	100.0	8	30.30	0.30	30.52	0.30	287	11	0.6

SK1	33301.1	YJB99.30	C:2	Basalt	Plag	315–500	30.08	0.26	0.3	100.0	26	30.00	0.30	30.04	0.40	301	39	0.3
WD7	33325.1	YJB99.38	F:2	Basalt	WR	350–500	*No plateau*				22	23.33	0.12	28.66	0.19	*200*	*2*	811.5
WD1	33323.1	YJB99.41	F:4	Basalt	HF'd WR	350–500	*No plateau*				16	26.64	0.16	30.64	0.18	*166*	*2*	35.2
WD1	33327.1	YJB99.41	F:2	Basalt	WR	350–500	*No plateau*				21	26.84	0.11	31.42	0.20	*157*	*2*	70.1

b

Sample	$^{40}Ar/^{39}Ar$ plateau age	$\pm 2\sigma$	Unit	Site
J56	27.07	0.13	Rhyolite lava	*
YJB99.87	27.67	0.12	Bayt Mawjan ig.	BM6
YJB88.79	29.48	0.13	Iftar Alkalb tuff	BB7
YJB99.76	29.59	0.12	Green tuff	BB5
YJB99.70	*No age*		Pumaceous tuff	BB2
YJB99.98	*No age*		Escarpment ig.	ESC4
YJB99.72	29.61	0.08	Akraban trachyan.	BB1
YJB99.23	30.30	0.14	Basalt	SK10
YJB99.13	30.03	0.26	Kura'a basalt	SK7
YJB99.46	29.94	0.24	Basalt	JS6
YJB99.26	30.51	0.36	Basalt	SK6
YJB99.33	30.20	0.26	Basalt	SK4
YJB99.30	30.24	0.18	Basalt	SK1
YJB99.38	*No age*		Basalt	WD7
YJB99.41	*No age*		Basalt	WD1

a) Ages are relative to Fish Canyon sanidine with an age of 28.02 Ma [16].
Errors shown are internal errors and reflect uncertainties in isotope measurements, corrections for reactor-induced interference and neutron fluence (J) determination. Ages calculated using the following decay constants: $\lambda(^{40}K)_{\varepsilon\chi} = 5.81 \times 10^{-11} \pm 3.4 \times 10^{-12}$ yr^{-1} (value from Steiger and Jager [61] using errors from [62]), $\lambda(^{40}K)_{\beta-} = 4.96 \times 10^{-10} \pm 1.7 \times 10^{-11}$ yr^{-1} [63], $\lambda(^{37}Ar) = 1.983 \times 10^{-2} \pm 4.4 \times 10^{-5}$ d^{-1} [64]. Nucleogenic correction ratios for reactor produced Ar isotopes from K and Ca were as in Knight et al. [17]. Plateau ages defined as in text. Italicized plateau ages are not accepted, and are shown for reference, only. Full analytical are available in the on-line supplement. Ages for units with multiple analyses are calculated using the weighted mean of all accepted plateau ages. The unit marked with asterisk (*) is sample J56, a rhyolite lava from southwestern Yemen (14.188°N; 44.272°E). b) Summary of new $^{40}Ar/^{39}Ar$ plateau ages presented in this paper. Ages for samples with multiple analyses are calculated using the weighted mean of all accepted plateau ages for the sample.

powered drill, and oriented using both magnetic and sun compass. The varying local magnetic field makes magnetic compass readings less reliable and only sun compass orientations are used in our analysis. In the laboratory the cores were sub-sampled into standard cylindrical specimens (in general 1–3 specimens per core). The structural attitudes of sampled units were estimated in the field and are given in Table 1 together with geographic coordinates of the sample sites. A total of 15 sites best representing the eruptive stratigraphy were selected for $^{40}Ar/^{39}Ar$ geochronology. The geographic locations of the sampling profiles are shown in Fig. 1C.

3. Analytical methods

3.1. Paleomagnetic demagnetization experiments

One specimen from each core was subjected to detailed demagnetization experiments at the paleomagnetic laboratories at Lund University and University of California, Santa Cruz. The majority of samples (~90%) were subjected to alternating field (AF) demagnetization using an automatic 3-axis AF demagnetizer, and the remaining ~10% were subjected to thermal cleaning using a laboratory-built furnace. Susceptibility versus temperature experiments indicate that the magnetic mineralogy is dominated by variably oxidized titanomagnetite for both basaltic and rhyolitic units, consistent with median destructive fields being lower than 40 mT and maximum unblocking temperatures ranging between 200 and 580 °C. For roughly 10% of the samples it was not possible to isolate the characteristic remanent magnetization (ChRM) due to strong lightning induced overprints indicated by stronger than average natural remanent magnetization (NRM) intensities and highly scattered within-site NRM directions. For two sites (JS5 and JK11) it was not possible to define a site–mean ChRM. For the remaining samples the direction of the ChRM was defined using standard principal component analysis [15] and we accept ChRM directions that could be defined with maximum angle of deviation less than 3° and the number of demagnetization points larger than 5. Typical examples of accepted samples are shown in Fig. 3. The site–mean directions are given in Table 1 and shown in geographic and stratigraphic coordinates in Fig. 4.

3.2. $^{40}Ar/^{39}Ar$ geochronology

A total of 15 sites best representing the eruptive stratigraphy was selected for $^{40}Ar/^{39}Ar$ geochronology. The samples were crushed and sieved into two or three fractions (generally 160–315, 315–500 μm, and as applicable, >500 μm). Matrix, mafic phenocrysts and feldpsar with inclusions were removed by use of a magnetic separator. Plagioclase separates were washed in purified water and alkali feldspar were bathed briefly in dilute (<1%) HF and then washed in purified water. Then ~20–30 mg of transparent feldspar was hand picked under a binocular microscope and loaded into a total of four aluminum disks. Two fine-grained basaltic samples (YJB99.38 and YJB99.41) contained only very tiny plagioclase ($\ll$63 μm), and were crushed, instead, for whole rock analysis. For these two samples 350–500 μm size fractions of cleaned whole rock matrix were washed in purified water. One whole rock sample (YJB99.41) was additionally prepared using a brief (~5 min) 5% HF acid leaching, and then washed. The three whole rock matrix samples (~50 mg of each) were then loaded into an additional aluminum disk.

Samples in each of the 5 aluminum disks, were bracketed with the monitor mineral Fish Canyon sanidine (FCS; assigned age of 28.02 Ma [16]) and co-irradiated for 5 h in the CLICIT facility of the TRIGA reactor at Oregon State University. After irradiation, ~10 mg of feldspar was degassed in 6–26 steps depending on sample behavior, using a defocused CO_2 laser (Synrad) and integrator lens, and analyzed with a MAP 215C mass spectrometer at the Berkeley Geochronology Center (BGC). Nucleogenic production ratios ($\pm 2\sigma$ errors) used to correct samples for reactor produced Ar isotopes from K and Ca were $(^{40}Ar/^{39}Ar)_K = (7.30 \pm 0.18) \times 10^{-4}$, $(^{39}Ar/^{37}Ar)_{Ca} = 6.95 \pm 0.18) \times 10^{-4}$ and $(^{36}Ar/^{37}Ar)_{Ca} = (2.67 \pm 0.14) \times 10^{-4}$ $^{37}Ar)_{Ca} = (2.67 \pm 0.14) \times 10^{-4}$ [17]. Plateau ages are defined as three or more contiguous steps corresponding to a minimum of 50% of the total ^{39}Ar gas released showing no significant slope and with the individual step ages agreeing within 2σ errors of the weighted mean age of the plateau segment. Only pla-

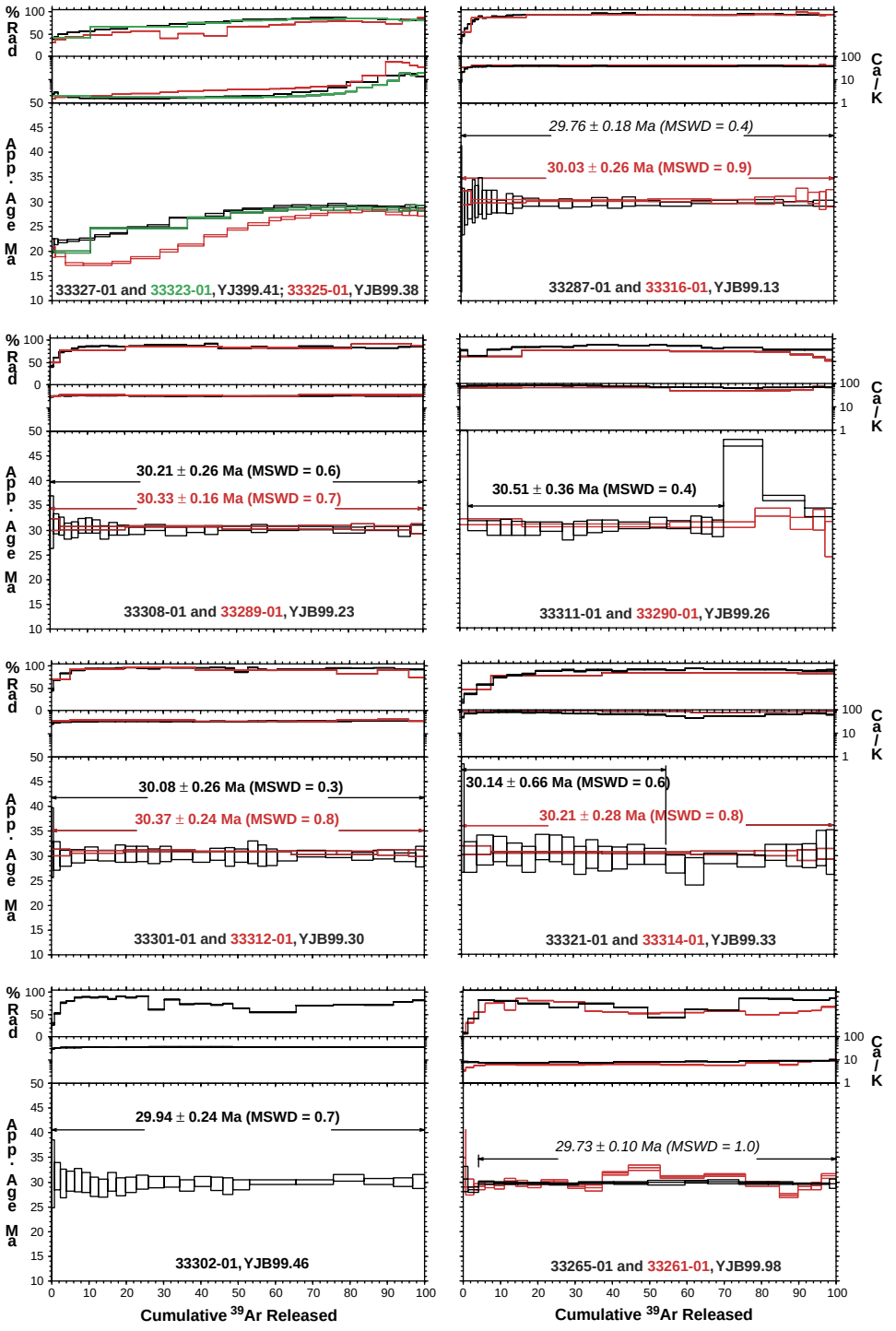


Fig. 5. $^{40}Ar/^{39}Ar$ experiments on mineral separates and whole rock samples from Yemen Oligocene volcanics as a function of total ^{39}Ar gas released. Analysis steps are shown with 2σ error bars. Plateau ages are also shown with 2σ errors, incorporating corrections from backgrounds, discrimination, decay, irradiation interferences, and include error on J (neutron fluence). Ages shown in italics are not accepted as discussed in the text. Ca/K and % radiogenic ^{40}Ar gas are also shown. Further details in Table 2.

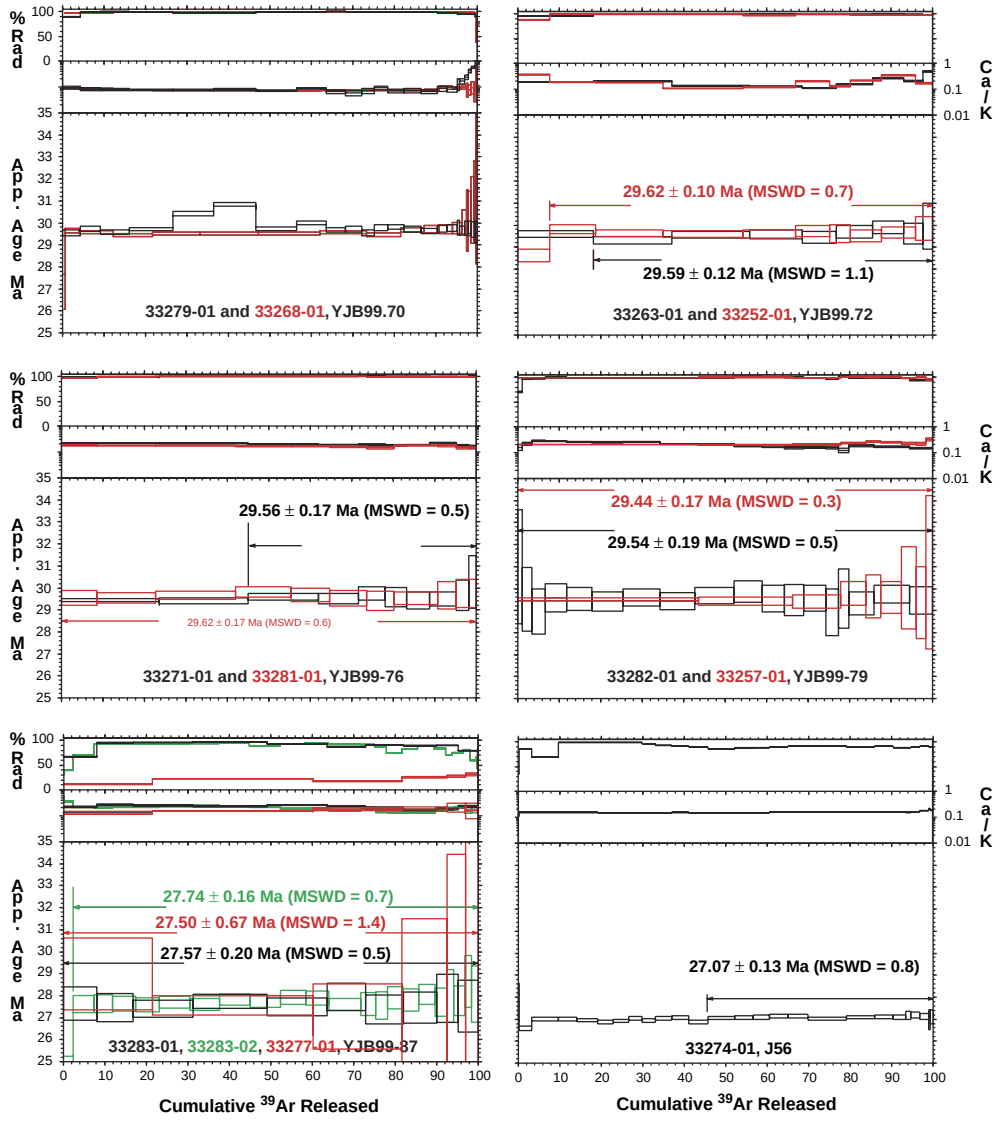


Fig. 5 (*continued*).

teau ages with MSWD (mean square of weighted deviates) ~1 are accepted, to ensure that data are not significantly scattered in excess of analytical error (MSWD > 1) or that analytical errors have not been overestimated (MSWD < 1).

Out of the 28 individual $^{40}Ar/^{39}Ar$ experiments (15 cooling units) a total of 21 experiments (11 cooling units) yielded acceptable plateau ages according to the above criteria. MSWD values for these accepted plateau age data range from 0.3 to 1.4, with a tendency towards overestimation of errors (MSWD < 1), likely attributable to correlation of error sources in the data regression process. Thus, on average the analytical uncertainties (i.e., reproducibility) of plateau ages reported herein may actually be slightly better than indicated by the errors we report. The individual sample analyses and unit mean-ages are summarized in Table 2, and discussed in detail below. Sample age spectra are shown in Fig. 5, while the full analytical data corrected for back-

grounds, mass discrimination and radioactive decay are available in the on-line data. Sample errors include internal errors, only, which reflect uncertainties in isotope measurements, corrections for reactor-induced interferences and neutron fluence (J) determination. We do not include systematic error such as absolute errors on decay constants or from determinations of the isotopic composition (age) of the standard. All previously reported $^{40}Ar/^{39}Ar$ plateau ages [13] from cooling units not dated in the present study and all related timescale data have been normalized to the same standard used in this study, e.g., the FCS monitor with an age of 28.02 Ma [16]. Recalculated $^{40}Ar/^{39}Ar$ plateau ages, are provided in Table 1 and all related timescale data [6] and ages determined in this study (Table 2b).

4. Paleomagnetic results

4.1. Magnetostratigraphy

Several cooling units were sampled along multiple profiles (see Fig. 2). The similar paleomagnetic site–mean directions of these duplicate sites (Table 1a) strongly indicate that they record the same time-spot of the paleomagnetic paleosecular variation, and confirm the independent volcanostratigraphic and geochemical correlation of the units. For the following analysis the site mean directions of duplicate sites are joined into cooling unit mean directions (Table 1b), and from the altogether 69 site–mean ChRM directions (Table 1a) we obtain 50 independent cooling unit means. We note that all consecutive lavas were

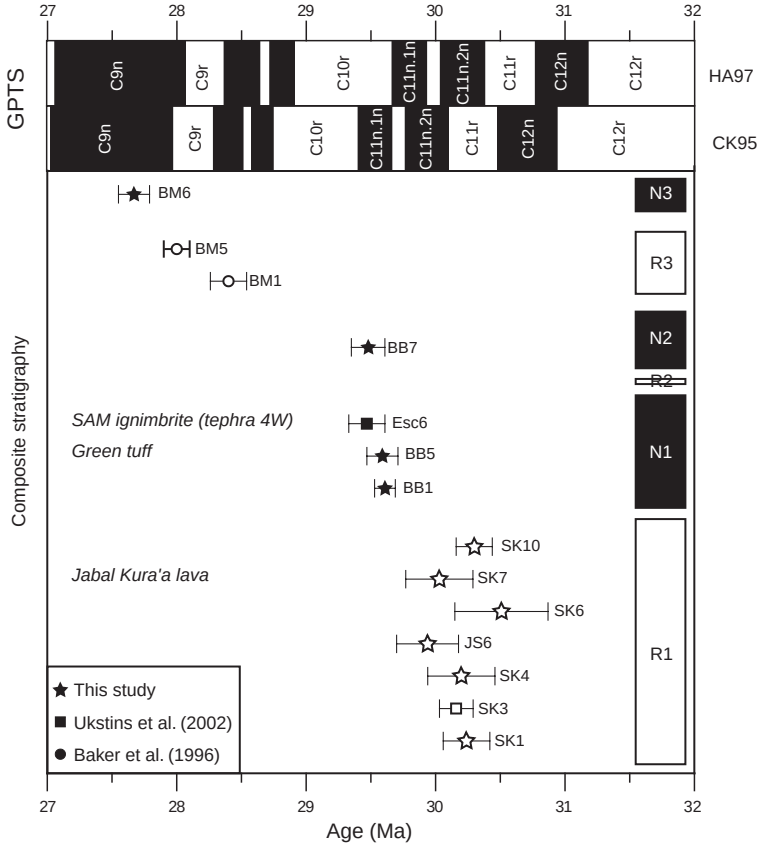


Fig. 6. Comparison of $^{40}Ar/^{39}Ar$ ages obtained in this study as well as those accepted from Baker et al. [13] and Utskins et al. [6] alongside the magnetostratigraphy of the Yemeni sections using the geomagnetic polarity scales of Cande and Kent [48] and Huestis and Acton [49]. Open/Solid symbols represent ages from reversed/normal polarity units. Section abbreviations are the same as for Fig. 2. See text for discussion.

found to record statistically different paleomagnetic directions, suggesting that sufficient time elapsed between the units to record different paleosecular variation, and by implication, a mode of emplacement slower than observed in other LIP provinces (e.g., [18,19]).

The composite stratigraphy contains five magnetic polarity zones, named R1–N1–R2–N2–R3 (Fig. 2). The five polarity zones are shown on Fig. 6 together with accepted $^{40}Ar/^{39}Ar$ ages from this study, recalibrated ages from the study of Baker et al. [13], and ages obtained from correlative units in Ethiopia [6]. Correlation of the Yemen polarity zones to the geomagnetic polarity time scale is discussed in Section 6.4.

4.2. Paleomagnetic pole and paleosecular variation

Two sites, SK1 and SK4, lie outside the 30.5° cut off angle defined by the method of Vandamme [20], hence recording excursional directions (see Table 1 and Fig. 7), and these sites are consequently excluded from the following analysis. Based on the remaining 48 independent cooling units we obtain a paleomagnetic pole with coordinates $\lambda = 74.2°$N, $\varphi = 249.1°$E (A95 = 3.6°). The angle between the mean normal (Dec = 356.2°, Inc = 7.1°, $\alpha_{95} = 8.4°$, $N = 15$) and reverse (Dec = 172.7°, Inc = 4.8°, $\alpha_{95} = 7.5°$, $N = 33$) polarity directions 9.2° is less than the critical angle 10.5° for which the two directional data set would be different at the 95% significance level and the collection therefore passes the reversal test at level C [21]. We estimate the paleosecular variation (PSV) as the angular standard deviation (ASD) of the virtual geomagnetic pole (VGP) distribution. Following McFadden et al. [22], ASD was corrected for within-site dispersion, Sw = 4.4°, and found to be 14.2° with 95% confidence limit $+2.3°/-1.7°$ [23]. This value of ASD is in good agreement with the value predicted (~15°) by the PSV model of McFadden et

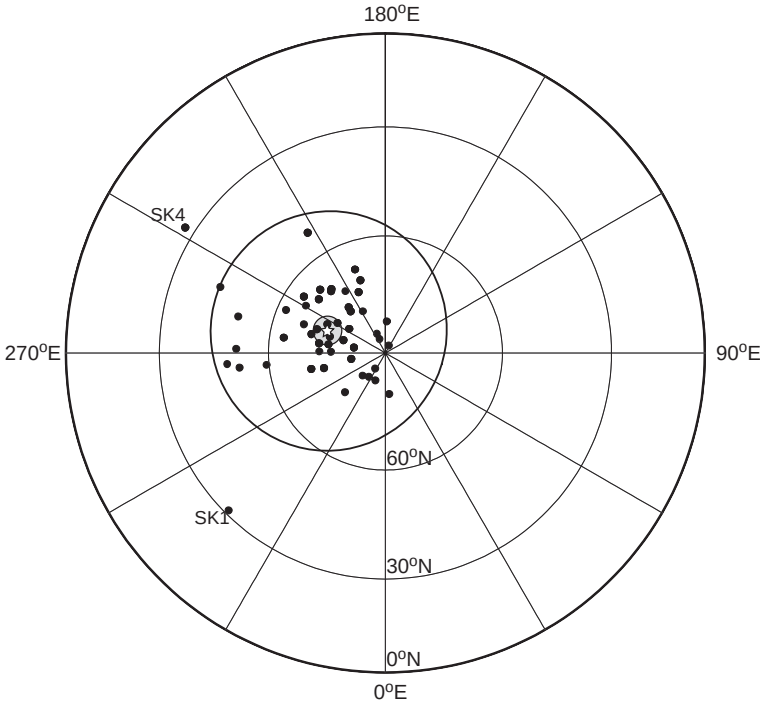


Fig. 7. Equal-area diagram of the 50 independent VGPs and the 30.5° cut off angle defined using the method of Vandamme [20]. Two sites from the Shibam Kawkabam profile (SK1 and SK4) lie outside the cut-off angle suggesting that they record excursional directions, and these sites are therefore not included in the paleomagnetic pole and paleosecular variation analysis. The mean pole (open star) and its 95% confidence cone is also shown.

al. [22] for the given age and paleolatitude, suggesting that secular variation has been sufficiently sampled and the mean VGP, i.e., the paleomagnetic pole, represents a time-averaged field.

4.3. Tectonic correction

Precise in-field estimates of the structural attitude are often difficult to obtain due to limitations in exposure, and the lack of the third dimension into the exposed rock surface. Since incorrect tectonic corrections may bias a paleomagnetic pole without affecting its statistical confidence limits it is important to recognize uncertainties related to tectonic correction. This is particularly true for apparently well-defined poles, such as the one obtained in present study, used for fine-scale paleogeographic and paleogeodynamo studies with the assumption that the A95 value represents the actual uncertainty. Comparison of in-field dip/strike measurements with highly precise multi-model stereo photogrammetry data in geological settings similar to Yemen of exposed and slightly dipping lavas has found that in-field dip/strike readings typically are 2–5° wrong [18]. It is therefore fair to assume similar uncertainties in our in-field dip/strike readings. Since several units were sampled along different profiles with variable structural attitude (e.g., Fig. 2 and Table 1), we may, in fact, test the in-field dip/strike estimates by comparing site mean direction prior to and after tectonic correction, i.e., by performing a fold test. Analyzing the entire dataset we observe a better grouping of data after tectonic correction (Fig. 4B) and also obtain a positive fold test [24] with the maximum untilting at 115% as demonstrated on Fig. 8. On the other hand, if we analyze only the SAM and Escarpment ignimbrites, the two units sampled at the greatest number of distinct locations (five), we find that the maximum in the precision parameters occurs at 80% (Sam ignimbrite) and 45% (Escarpment ignimbrite) untilting. This suggests that dip values may have been slightly underestimated (Fig. 8), or alternatively that there is some noise in the paleomagnetic recording (e.g., [25]). In either case, the analysis underlines that the actual uncertainty of the site–mean directions are larger than suggested by their α_{95}-values. Given that the structural attitudes are different for different profiles

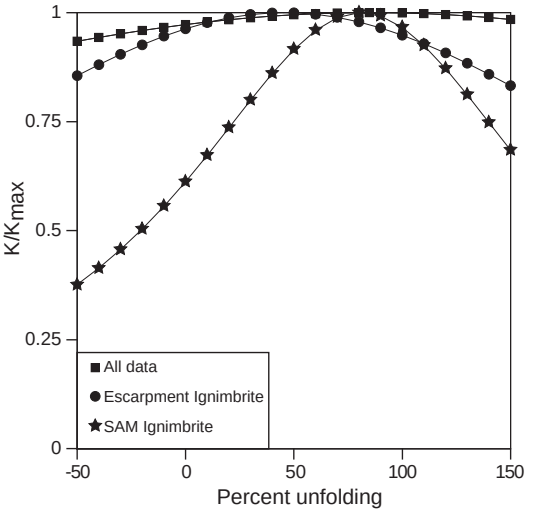


Fig. 8. Fold test [24] of the 50 independent cooling units (Table 1) is positive with the maximum untilting at 115%. Fold test on the SAM and Escarpment ignimbrites (5 different locations) shows maximum untilting at 80% and 45%, suggesting that dip values may have been slightly underestimated.

(Table 1a) the uncertainties related to in-field strike/dip estimates will tend to average out, but nevertheless the 3.6° A95-value is an underestimate of the actual uncertainty.

5. $^{40}Ar/^{39}Ar$-dating results

Results are presented in Table 2a. The lowest basalt flows were not datable due to lack of datable single mineral phases and problematic ^{39}Ar recoil induced by irradiation of the fine-grained whole rock matrix. Despite a lack of data from the lowest units, accepted ages from the lower mafic units yielding reliable results span 30.51 ± 0.36 to 29.94 ± 0.24 Ma, a statistically unresolvable interval. The total duration of silicic volcanism measured in this study (Table 2b) spans the earliest silicic volcanism as early as 29.7 ± 0.1 Ma to the latest sampled silicic volcanism around 27.1 ± 0.1 Ma. These silicic unit ages bracket ~3 Ma, a considerably longer duration than that implied by the underlying basaltic volcanism. Evidence for episodic volcanic behavior is suggested by both $^{40}Ar/^{39}Ar$ studies (details below in stratigraphic order from youngest to oldest) and paleomagnetic data. All $^{40}Ar/^{39}Ar$ age data (this

study and other studies) are presented (recalculated, unless otherwise stated) relative to an age of 28.02 Ma for Fish Canyon sanidine.

5.1. Rhyolite lava (J56)

This sample is the same alkali feldspar separate previously dated by Baker et al. [13], analyzed in 7 steps, yielding a plateau age of 26.67 ± 0.12 Ma. Our data, comprising 26 degassing steps, show more complex behavior despite yielding an accepted plateau age (54.3% of total ^{39}Ar) of 27.07 ± 0.13 Ma with an MSWD of 0.8, and an atmospheric ^{40}Ar/^{36}Ar isochron intercept of 293 ± 10. Our determined age is older than that of Baker et al. [13], well outside of 2σ errors. Both samples show similar amounts of radiogenic ^{40}Ar, and have total gas ages in very good agreement with their plateau ages. Some discrepancy remains, which may be a result of differences in laboratory and data regression procedures. Sample comparisons between this study and the paper of Baker et al. [13] reveal systematically younger ages for the later study. While the best age for this unit remains unclear, the greater detail presented here may be a more realistic representation of the complexity of this sample.

5.2. Unit BM6 (Bayt Mawjan ignimbrite), sample YJB99.87

Alkali feldspar crystals were analyzed from 160–315 and 315–500 μm size fractions. A third analysis, duplicating the larger size fraction, was also performed with greater heating detail (26 steps). The smallest size fraction displays 10–30% radiogenic ^{40}Ar gas content, and was analyzed in six steps yielding a plateau with correspondingly larger error bars of 27.50 ± 0.67 Ma and a larger MSWD of 1.4. The larger grain size yields identical ages of 27.57 ± 0.20 and 27.74 ± 0.16 Ma, and atmospheric ^{40}Ar/^{36}Ar isochron intercepts (298 ± 17 and 300 ± 9, respectively), indistinguishable from the smaller grain size (intercept of 299 ± 4). The more detailed heating schedule reveals greater variation in apparent age and % radiogenic ^{40}Ar (80–90%) between individual steps, but results in an MSWD of 0.7 suggestive of less scatter than expected. A lower MSWD (0.5) is also seen in the less detailed heating schedule (10 steps) of the larger grain size. The weighted mean of these three accepted plateau ages is 27.67 ± 0.12 Ma.

Alkali feldspar separate (JB232) from this same unit sampled within the Bayt Mawjan section was also dated by Baker et al. [13] in 11 steps, yielding a plateau age of 27.91 ± 0.10 Ma. The age spectrum shows noticeable disturbance with the two final steps yielding higher apparent ages, possibly a result of excess Ar, and slightly younger apparent ages in the initial 4 steps. Our three new ages for this unit do not show similar signs of disturbance, and may be considered more robust.

5.3. Unit BB7 (Iftar Alkalb), sample YJB99.79

Two experiments on 160–315 and 315–500 μm fractions of alkali feldspar yield identical, well-behaved plateau ages of 29.44 ± 0.17 and 29.54 ± 0.19 Ma. Both accepted plateaus are chosen from 100% of ^{39}Ar gas released, and yield MSWDs of 0.3 and 0.5, respectively, as well as atmospheric ^{40}Ar/^{36}Ar isochron intercepts (311 ± 132 and 316 ± 54). The larger size fraction, with a slightly older age, shows inverse compositional zoning, with the Ca/K ratio changing from ~0.5 to ~0.1 over the duration of the heating experiment. The weighted mean of these two ages is 29.48 ± 0.13 Ma.

An alkali feldspar separate (sample JB163) from the Iftar Alkab unit sampled within the Jabal Kura'a section [13] was reported yielding a plateau age of 28.72 ± 0.16 Ma. This age is slightly younger than our age and shows some higher apparent age disturbance in the initial three steps, with the rest of the spectra forming a flat plateau (87.8% of total ^{39}Ar gas released). The chaotic nature of this tuff, the presence of inverse zoning in the larger grain size analysis of this study, and total the variation in these ages may indicate the presence of crystal inheritance (xenocrystic contamination). Single crystal analyses could potentially address this.

5.4. Unit BB5 (green tuff), sample YJB99.76

Two experiments on 315–500 and >500 μm alkali feldspar separates from this tuff yield two accepted plateau ages of 29.56 ± 0.17 and 29.62 ± 0.17 Ma, respectively, with MSWDs of 0.5 and 0.6. Both grain sizes show similar behavior with slightly youn-

ger initial steps and similar Ca/K ratios, and minor potassium enrichment in the highest heating step. Larger error bars, due to smaller sample size and correspondingly smaller analyzed signal, allows inclusion of all steps from the largest size fraction analysis, though these steps are not included in the similar 315–500 μm plateau. Both samples show excellent agreement between their total gas ages and plateau ages, and display atmospheric $^{40}Ar/^{36}Ar$ isochron intercepts (247 ± 229 and 264 ± 97). Their weighted mean age is 29.59 ± 0.12 Ma.

This unit was also sampled as sample JB276 from the Jabal Kura'a section [13] yielding a plateau age of 29.30 ± 0.12 Ma in 9 steps. The apparent age spectra shows slightly disturbed initial steps, followed by the plateau (3 steps), a younger step, and a final high temperature step yielding an apparent age of ~31 Ma, possibly a result of excess Ar.

5.5. Unit BB2 (Bayt Baws pumaceous tuff), sample YJB99.70

Alkali feldspar separates of two size fractions (160–315 and 315–500 μm) were analyzed. Both samples yield non-atmospheric $^{40}Ar/^{36}Ar$ isochron intercepts (264 ± 16 and 350 ± 12, respectively) although the amount of ^{36}Ar in several of the highest temperature steps was below detection limits. The larger grain size experiment shows a large jump in apparent age in the early heating steps. This is not directly related to a change in K/Ca, nor a change in % radiogenic ^{40}Ar gas. After this jump, the sample shows gradual enrichment towards Ca and variable apparent ages. A 'forced' plateau age consisting of the post-jump steps yields an age of 29.7 ± 0.2 Ma, with an MSWD of 3.7, indicative of an unacceptable amount of excess scatter. The apparent age spectrum of the smaller size fraction is slightly saddle shaped suggesting the redistribution of ^{40}Ar and mirrors the compositional data, showing an apparent age increase coincident with a shift towards more calcic compositions. A 'forced' plateau through the finer fraction data including 88% of the gas (all but the first three steps) yields an age of 29.55 Ma ± 0.14 with an MSWD of 3, clearly indicating excess scatter. Integrated ages from these two samples do not agree within 2σ errors, ranging from 29.61 ± 0.14 Ma in the finer grain size to 29.89 ± 0.14 Ma in the larger grain size. No age is concluded from this unit.

5.6. Unit Esc4 (escarpment ignimbrite), sample YJB99.98

Two analyses of feldspar separate (160–315 and 315–500 μm fractions) yield one plateau age of 29.73 ± 0.10 Ma with an MSWD of 1.0 from the larger size fraction. This sample, heated in 15 steps, shows little compositional zoning (Ca/K ~8.5), but large shifts in % radiogenic ^{40}Ar gas, varying chaotically between 10% and 80%. The smaller size fraction, analyzed in 21 steps, shows a disturbed spectrum with older apparent ages in the middle heating steps and pronounced compositional zoning from Ca/K of ~2 to Ca/K of ~8.5. The total gas age of the smaller size fraction is 30.08 ± 0.16 Ma. This contrasts with the larger size fraction that yields a total gas age of 29.72 ± 0.12 Ma, in good agreement with the accepted plateau age. A more detailed step heating schedule, along with a non-atmospheric $^{40}Ar/^{36}Ar$ isochron intercept (307 ± 2) suggests the presence of excess ^{40}Ar in the smaller size fraction. Excess ^{40}Ar may also be present in the larger grain size fraction, where the plateau age realized is clearly older than robust ages concluded from underlying units, but is not obviously apparent (with a $^{40}Ar/^{36}Ar$ isochron intercept of 298 ± 3). No age is concluded for this unit.

5.7. Unit BB1 (Akraban trachyandesite), sample YJB99.72

Two alkali feldspar separates (160–315 and 315–500 μm fractions) yielded two accepted plateau ages of 29.59 ± 0.12 and 29.62 ± 0.10 Ma, respectively, with MSWD values of 1.1 and 0.7. These ages are identical within 2σ errors, and both analyses yield $^{40}Ar/^{36}Ar$ isochron intercepts (316 ± 57 and 294 ± 58) indistinguishable from the atmospheric value. The smaller size fraction shows a more pronounced compositional zoning, enriched in Ca towards higher temperatures with the Ca/K ratio changing from ~0.1 to ~0.5. The larger size fraction shows a constant Ca/K ratio of ~0.2 for the entire experiment. The weighted mean age of these samples is 29.61 ± 0.08 Ma.

5.8. Unit SK10, sample YJB99.23

Plagioclase separates (160–315 and 315–500 μm) from this basaltic lava yield two accepted plateau ages of 30.21 ± 0.26 and 30.33 ± 0.16 Ma, respectively, with MSWD values of 0.6 and 0.7. The larger grain size was heated in 8 steps, while the smaller grain size was heated in 25 steps. Both apparent age spectra show similar behavior with younger initial ages at the lowest temperature steps, typical of minor alteration, followed by fairly constant Ca/K ratio of ~35 and 80–90% of radiogenic ^{40}Ar throughout the rest of the experiment. ^{40}Ar/^{36}Ar isochron intercepts overlap atmospheric values (315 ± 20 and 298 ± 12, respectively) and show no obvious signs of excess ^{40}Ar. The weighted mean age of this lava flow is 30.30 ± 0.14 Ma.

5.9. Unit SK7 (Kura'a basalt), sample YJB99.13

Two step heating experiments on 160–315 and 315–500 μm plagioclase separates from this unit yield two plateau ages of 30.03 ± 0.26 and 29.76 ± 0.18 Ma, respectively, with MSWD values of 0.9 and 0.4. A detailed step heating of the larger grain size (26 steps) shows younger apparent ages at the lowest temperature steps and very uniform behavior thereafter, with a constant Ca/K of ~35 and ~95% of radiogenic ^{40}Ar. The small MSWD indicates less scatter than expected, however, and the age is statistically resolvable from that of the smaller grain size. The smaller grain size fraction, heated in 11 steps, shows very similar behavior with more scatter and slightly older ages in the highest temperature steps. There may be minor amounts of excess ^{40}Ar present in the smaller size fraction (with its slightly older age). The ^{40}Ar/^{36}Ar isochron intercepts, however, reflect atmospheric ratios for both experiments (300 ± 16 and 282 ± 30, respectively). The finer grain size yields the accepted age for this unit of 30.03 ± 0.26 Ma.

Plagioclase sampled from this unit in the Jabal Kura'a section (sample JB148) was previously dated [13] yielding an age of 29.40 ± 0.28 Ma. Results of the 8 step degassing experiment yielded older initial ages, a three step plateau containing 51.8% of the total ^{39}Ar gas released, and three younger ages. The reported age of this sample [13] is significantly younger than our determined age, fitting the general trend of slightly younger age determinations from the previous study.

5.10. Unit JS6, sample YJB99.46

Plagioclase separate (315–500 μm) from this basaltic lava was degassed in 26 steps yielding an accepted plateau age of 29.94 ± 0.24 Ma with an MSWD of 0.7. The step heating spectra shows constant apparent age, with minor K-rich zoning of initial temperature steps and variable radiogenic ^{40}Ar content from ~25% to ~85%. The ^{40}Ar/^{36}Ar isochron intercept (300 ± 8) reflects atmospheric ratios, and the total gas age and ^{40}Ar/^{36}Ar isochron age are in excellent agreement with the determined plateau age.

5.11. Unit SK6, sample YJB99.26

Two plagioclase separates (63–125 and 125–425 μm) were degassed from this basaltic lava. The smallest size fraction, degassed in 7 steps, yields an apparent age spectra with excess scatter in the high temperature steps correlated with a spike in the Ca/K ratio (from ~45 to ~65) and a drop in the radiogenic ^{40}Ar gas content. The sample shows a non-atmospheric ^{40}Ar/^{36}Ar isochron intercept (326 ± 23), and yields no plateau. The larger grain size, degassed in 21 steps, shows two initial steps with high apparent ages followed by 69% of total 39Ar gas release in a plateau with an age of 30.51 ± 0.36 Ma and a MSWD of 0.4, which is then interrupted by a sudden increase in apparent ages at the highest three temperature steps. This high temperature increase correlated with a drop in the Ca/K ratio of the sample, and is the likely result of degassing K-rich inclusions. The initial low temperature ages indicate the presence of excess ^{40}Ar, although an isochron yields an atmospheric ^{40}Ar/^{36}Ar intercept (295 ± 24). The accepted plateau age from the larger grain size is considered the age of the unit, 30.51 ± 0.36 Ma.

This unit was previously dated (MNY221) by whole rock analysis [13] in the Shibam–Kawkabam section. This analysis, in eight steps, shows recoil disturbance and did not yield an age, though an estimated (not plateau) age of 29.8 ± 0.2 Ma was reported. This estimated age is statistically distinct at 95% confidence from our new age determination, and should be considered less reliable.

5.12. Unit SK4, sample YJB99.33

Step heating experiments on plagioclase separates (160–315 and 315–500 μm size fractions) from this basaltic lava yield two plateau ages. The smaller grain size, heated in 7 steps, yields a plateau age of $30.21 \pm .28$ with an MSWD of 0.8. This grain size displays a higher (~80) Ca/K ratio. The larger grain size, heated in 25 steps, yields a plateau age of 30.14 ± 0.66 Ma with a MSWD of 0.6 that comprises the initial 55% of total ^{39}Ar gas released, only. The entire spectrum, however, overlapping within 2σ errors (larger due to the more detailed step heating schedule with less gas in individual steps), yields a 'forced' plateau age of 29.7 ± 0.5 Ma, demonstrating the effect of the slightly lower apparent ages at high temperature. The larger grain size also shows strong variations in Ca/K ratio from ~45 to ~80, with zoning becoming more calcic in the higher temperature steps. Both samples yield ^{40}Ar/^{36}Ar isochrons with atmospheric intercepts (303 ± 8 and 303 ± 14 for the smaller and larger size fractions, respectively) and isochron ages in excellent agreement with the determined plateau ages. The weighted mean age of these two accepted age plateaus is 30.20 ± 0.26 Ma.

5.13. Unit SK1, sample YJB99.30

Two analyses of plagioclase separates (160–315 and 315–500 μm size fractions) from this basaltic unit yield two plateau ages indistinguishable within 2σ errors. The smaller grain size, degassed in 8 steps, shows uniform behavior yielding a plateau age of 30.37 ± 0.11 Ma and MSWD of 0.8. The larger grain size, heated in 26 steps, also shows uniform behavior, yielding a plateau age of 30.08 ± 0.26 Ma with a much lower MSWD of 0.3 suggesting possible overestimation of analytical errors. Both experiments yield isochrons with atmospheric ^{40}Ar/^{36}Ar intercepts (287 ± 11 and 301 ± 39 for the smaller and larger size fractions, respectively) and consistent total gas ages and isochron ages. The weighted mean average age for these two accepted plateau ages is 30.24 ± 0.18 Ma.

5.14. Unit WD7, sample YJB99.38

Fresh appearing whole rock matrix from this basaltic lava was degassed in 22 steps yielding a highly discordant age spectrum suggestive of ^{39}Ar recoil redistribution effects. Initial apparent ages decrease from ~20 Ma before increasing steadily to a maximum apparent age of ~28 Ma at the highest temperature steps. Ca/K ratios begin low (<10), climbing steadily until the highest temperature steps, where they jump sharply to ~60, and then decrease. This is suggestive of multiple phases degassing at various temperature steps combined with minor alteration, as well as a strong effect from ^{39}Ar recoil due to the fine-grained nature of the whole rock matrix. An isochron yields a sub-atmospheric ^{40}Ar/^{36}Ar intercept of 200 ± 2.

5.15. Unit WD1, sample YJB99.41

One sample from this basaltic lava was analyzed in two experiments, one with no treatment and one that had been briefly leached with 10% HF acid. HF rinsing is a commonly used method for preferentially dissolving fine-grained and often more soluble alteration products [26]. Both samples show strong ^{39}Ar recoil effects, with the maximum apparent age for both samples achieving ~29 Ma. This may be considered a minimum age for these samples but provides no further age information. Both samples yield isochrons with sub-atmopsheric ^{40}Ar/^{36}Ar intercepts of 157 ± 2 and 166 ± 2 (HF'd sample). Both samples also show a gradual increase of Ca/K content and little evidence for alteration, although data from the sample that underwent HF leaching show noticeably less scatter when plotted as an isochron, as well as less scatter in the highest temperature steps.

6. Discussion

6.1. Paleomagnetic poles and paleogeography of the Afro-Arabian traps

The Ethiopian part of the Afro-Arabian traps has been extensively studied, and all available paleomagnetic data were recently analyzed by Kidane et al. [27]. They suggest that only data based on detailed demagnetization and principal component analysis should be considered. Combining all reliable paleomagnetic data they obtain a well-defined pole for the Ethiopian part of the province based on 76 independent paleomagnetic units (see Table 3).

Table 3
Age is the reported age of the paleomagnetic or APWP pole; Plat, Plong are the latitude and longitude of the paleomagnetic pole; A95 is the confidence cone; N is the number of independent readings of the paleomagnetic field included in pole determination

Paleomagnetic pole							Finite rotation poles onto Africa			Rotated pole	
Study	Age (Ma)	Plat (°N)	Plong (°E)	A_{95} (°)	N	Reference	λ (°N)	φ (°E)	ω (°)	Plat (°N)	Plong (°E)
1. Yemen	29 ± 2	74.2	249.1	3.6	48	This study	32.0	25.0	−7.75	77.6	222.1
2. Ethiopia	30 ± 1	78.7	209.4	3.4	76	[27]	0	0	0	78.7	209.3
3. Kerguelen	28 ± 2	85.8	197.7	4.0	94	[66]	14.7	310.4	4.60	81.5	210.0
4. W. Greenland	35 ± 5	76.3	201.5	7.4	13	[67]	76.9	7.3	8.30	75.6	216.5
5. Australia	30 ± 5	68.9	272.4	4.3	52	[68]	13.8	48.4	−17.40	74.3	217.5
6. Germany	39 ± 5	80.8	182.0	4.2	47	[69]	21.0	337.6	4.25	78.5	201.4

The finite rotation used to transfer paleomagnetic poles into African coordinates are derived from [65], except for the Yemen pole that is transferred using the rotation pole of [33].

The Arabian part of the Afro-Arabian traps has been less studied and only one paleomagnetic pole exists, from the As Sarat volcanic field in southwestern Saudi Arabia [28]. The failed reversal test for the As Sarat pole was originally taken to support a displaced dipole source of the Oligocene Earth's magnetic field [28] but this hypothesis is not supported by our data nor by the Ethiopian data [27] and it is possible that the failed reversal test is related to the crude demagnetization technique, which did not remove secondary overprints. The As Sarat paleomagnetic pole lies 4.6° away from the pole obtained in the present study and is statistically distinct at the 95% confidence level. Therefore we exclude the As Sarat paleomagnetic data [28] from the following analysis.

6.2. Oligocene Afro-Arabian paleogeography

Seafloor magnetic anomalies and geochronology of dredged basaltic rocks indicate that sea-floor spreading has been taking place only since 5–6 Ma in the Red Sea [29]), while significant pre-drift extension is believed to have taken place since ~25 Ma [10]. The amount of pre-drift extension is difficult to quantify, however, and it varies between different plate kinematic models that aim to restore the Afro-Arabian paleogeography prior to rifting (e.g., [29–35]). Based on our Yemen paleomagnetic pole and that of Ethiopia [27] we may attempt to test the various plate kinematic models in a manner similar to analyses of pre-drift extension at Atlantic bordering continents [36,37]. Although the A95 values of ~3° for the Ethiopian and Yemen paleomagnetic poles are too large to differentiate between all the kinematic models with statistical significance, we note that the plate kinematic model of Joffe and Garfunkel [33] best brings the poles in agreement (Table 3 and Fig. 9). This plate kinematic finite rotation pole assumes a pre-drift extension of ~200 km in the Red Sea prior to the initiation of centralized sea floor spreading, slightly more than other models.

6.3. Geometry of the Oligocene time-averaged paleomagnetic field

A fundamental concept in paleomagnetism is the so-called geocentric axial dipole (GAD) hypothesis, which states that the Earth's magnetic field is, on the average, a GAD. Assuming the GAD hypothesis is correct, paleomagnetic poles yield quantitative estimates of both rotations around vertical axes and latitudinal tectonic transport. The GAD hypothesis is therefore very important for paleomagnetic paleogeographic reconstructions. For the last 5 million years, where plate tectonic motion can be ignored and tests of the GAD hypothesis are relatively simple, it appears that the time-averaged field configuration largely matches that of a GAD, although a small ~5% quadrupole contribution cannot be excluded [38,39]. For older intervals in the Earth's history, paleomagnetic data are scarce and plate tectonic reconstructions are less certain. It is therefore more difficult to test the GAD hypothesis, which may explain why some authors have argued for significant long term non-dipole fields [4,36,40] while other studies suggest a GAD field [41,42].

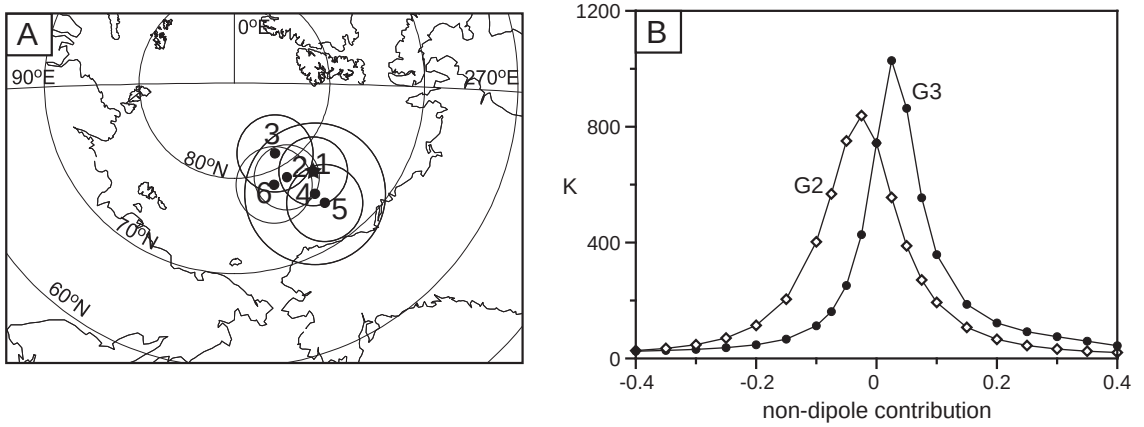


Fig. 9. (A) Equal-area projection of the Yemen paleomagnetic pole (star) rotated to Africa together with contemporaneous poles from volcanic lava sequences (see Table 3); all poles are in African coordinates. (B) Normalized precision parameter (k) for the paleomagnetic poles listed in Table 3 assuming various contributions of a quadropole (G2) and octupole (G3) relative to the dipole.

Based on the new Oligocene paleomagnetic pole presented here as well as selected contemporaneous poles (Table 3), we test whether the Oligocene paleomagnetic field differed significantly from a GAD as suggested in some studies [4,43,44]. To avoid uncertainties related to inclination shallowing in sedimentary paleomagnetic data [45,46] and the lack of tectonic control in intrusive magmatic units [47] all selected poles stem from lava sequences. Poles are calculated from at least 10 sites, with more than five samples per site and each site having Fisher dispersion parameters suggesting that paleosecular variation has been sampled and averaged out. The six paleomagnetic poles listed in Table 3 are the only poles in the 25–35 Ma time window fulfilling these selection criteria. We examine potential non-dipole contributions to the field by calculating the paleocolatitudes (and the paleomagnetic poles) for the six studies listed in Table 3 for their various non-dipole contributions. The observed inclination (I) is related to the expected paleocolatitude (θ) by:

$$\tan(I) = \frac{2\cdot\cos\theta + G2(4.5\cdot\cos^2\theta - 1.5)}{\sin\theta + G2(3\cdot\cos\theta\sin\theta)}$$

where G2 is the quadrupole/dipole field ratio

$$\tan(I) = \frac{2\cdot\cos\theta + G3(10\cdot\cos^3\theta - 6\cdot\cos\theta)}{\sin\theta + G3(7.5\cdot\cos^2\theta\sin\theta - 1.5\cdot\sin\theta)}$$

where G3 is the octupole/dipole field ratio.

After solving for θ by an iterative method the resulting poles are transferred into African coordinates by the relative rotation poles listed in Table 3 and analyzed in pole space. As shown on Fig. 9A, the poles group well for a purely GAD field. In Fig. 9B the precision parameter, K, is given for various G2 and G3 values and a slightly better grouping of poles, i.e., a slightly higher K-value is found for an octupole contribution G3 of 2.5% or an quadrupole G2 contribution of −2.5% (Fig. 9B). A suggested quadrupole G2 term of 14% as suggested by Rochette et al. [4] is not allowed by our analysis. An octupole contribution larger than 6% [36,44] is also not supported. In fact, our analyses suggest that the Oligocene time-averaged field does not require more of a non-dipolar contribution than that indicated by analysis of the 0–5 Ma paleomagnetic database [38,39].

6.4. Afro-Arabian flood volcanism and the Oligocene gpts

Before attempting to correlate our Yemeni polarity zones and $^{40}Ar/^{39}Ar$ ages to the Oligocene geomagnetic polarity time scale (GPTS) it is necessary to consider possible GPTS age uncertainties. The two Oligocene GPTS used in the present study, namely the ones of Cande and Kent [48] (CK95) and Huestis and Acton [49] (HA97), are based on the same geochronologic 'tie-points' but nevertheless differ ~0.3 Ma (e.g., Fig. 6) because they use different marine mag-

netic-anomaly data. Considering also uncertainties of the geochronologic 'tie-points' used to calibrate the GPTS, which are the same for both the CK95 and HA97, it is fair to suspect that the age uncertainties may be larger than suggested by the ~0.3 Ma age difference between CK95 and HA97. The first 'tie-point' of 23.8 Ma for the onset of 6Cn.2r is based on a chronogram from Berggren et al. [50]. Chronograms are statistical compilations of age data often determined from a combination of multiple radioisotopic dating systems. Considering the resolution possible with modern methods they are largely an obsolete or last-resort method for determining boundary ages, but have formed the basis for many absolute age correlations between the magnetic and absolute timescale. Astronomical tuning of sedimentary records [51] suggest that the 6Cn.2r 'tie-point' may be 0.9 Ma too old. The second 'tie-point' of 33.7 Ma for the 0.14 portion of C13r [48] is derived from a combination of radioisotopic ages including Rb–Sr, K–Ar and $^{40}Ar/^{39}Ar$ work [52], whose intercalibration is uncertain at the 1% level or more [53] and continues to demand active revision (e.g., [54]). Also worth considering is the suggested correlation [55] of the CK95 (and therefore the HA97) GPTS to the $^{40}Ar/^{39}Ar$ dating standard FCS with an assigned age of 27.84 Ma, although this standard was not directly involved in the determinations of the tie-points and would be relevant only to $^{40}Ar/^{39}Ar$ age determinations. Normalizing the CK95 and HA97 GPTS to the FCS monitor age of 28.02 Ma [16] that is used in this study shifts the GPTS polarity boundaries ~0.2 Ma towards older ages (see online dataset). Realistically, an accurate comparison between timescales and unit ages presented here can only be made if the tie-points used for these timescales are re-dated by modern comparable $^{40}Ar/^{39}Ar$ methods. Taken altogether, GPTS Oligocene age uncertainties could easily be as large as the typical ~0.5 Ma length of early Oligocene magnetochrons, and it is therefore difficult to uniquely correlate the Yemeni polarity zones to the GPTS based solely on the $^{40}Ar/^{39}Ar$ geochronology. A means to circumvent radioisotopic age uncertainties is given by the study of Ukstins Peate et al. [56] that, based on Pb and Nd isotope ratios and rare earth element compositions, demonstrated that the Jabal Kura'a and SAM ignimbrites (both part of the N1 normal, see Fig. 2) are the source of ash layers identified in ocean drilling program (ODP) drill cores at site 711. Since these ash layers are correlated to magnetochron C11n based on biostratigraphy data [57] it appears that central Yemeni N1–R2–N2 polarity sequence corresponds to magnetochrons C11n.2n–C11n.1r–C11n.1n, which is, in fact, in good accordance with the $^{40}Ar/^{39}Ar$ and the CK95 time scale (e.g., Fig. 6). The underlying R1 polarity zone therefore should correspond to C11r, which is also supported by six $^{40}Ar/^{39}Ar$ age data (this study) giving a mean age of 30.19 ± 0.20 Ma that correlates well with the CK95 age of C11r but less well with the HA97 time scale. An onset of volcanism in C11r in Yemen is similar to Ethiopian magnetostratigraphy [4,5], and, thus supports the hypothesis of a coeval onset Oligocene flood volcanism throughout the Afro-Arabian LIP.

The R3 reversed polarity section is constrained by two ages on silicic units of 28.35 ± 0.14 and 27.92 ± 0.10 Ma (JB228II and JB232, respectively, [13]), and has an estimated duration of $\sim 0.5 \pm 0.1$ Ma. The N3 normal polarity zone is represented by a single unit, the Bayt Mawjan ignimbrite with an age of 27.67 ± 0.12 Ma. Thus the transition from R3 to N3 is estimated to have occurred at $\sim 27.8 \pm 0.2$ Ma. While it is difficult to uniquely correlate the R3 polarity zone to the GPTS, the overlying and youngest polarity zone, N3, appears to share good correspondence with C9n.

The onset of the N2 normal polarity period is well constrained by the age of the Iftar Alkalb flow (29.48 ± 0.13 Ma), but this may be succeeded by an eruptive hiatus. None of the subsequent overlying units in the N2 period are constrained by accepted age data, and thus the period of time represented by N2 could be as long as 1 Ma. The R2 polarity zone, which, based on strong outside constraints [56,57], corresponds to the brief C11n.1r Subchron, has a well-constrained age of $\sim 29.5 \pm 0.1$ Ma (based on the ages 29.47 ± 0.14 Ma age of the underlying SAM ignimbrite and 29.48 ± 0.13 Ma for the overlying BB7 unit), potentially providing a new tie-point for calibration of the GPTS. However, an $\sim 29.5 \pm 0.1$ Ma age for C11n.1r is only in marginal agreement with previous determinations. Coulie et al. [7] define the onset of C11n.1r to have taken place at 29.75 ± 0.10 Ma. Unfortunately, these reported $^{40}Ar/^{39}Ar$ ages [7] are based on a standard (HD-B1) whose intercalibration with our standard is not well established, and an

accurate comparison of ages between this study and that of Coulie et al. [7] cannot be made at this time. Additional studies provide $^{40}Ar/^{39}Ar$ and magnetic data from the Ethiopian part of the Afro-Arabian traps [4,5,57] assigning an age of 30.1 ± 0.1 Ma to the underlying C11n.2n, in good agreement with our data, but a slightly older age of 29.7 ± 0.1 Ma to the end of C11n.1r. The reason for the discrepancy in the age of the brief C11n.1r Subchron remains unclear at this time. One possible explanation is that the R2 reverse polarity zone is an excursion within C11n, and that the actual C11n.1r is not recorded in our sampled sequences.

Underlying the brief R2 reversal is a set of normal polarity lavas and pyroclastic units within the N1 zone. The reverse to normal transition is bracketed by ages of the upper basalts (at the top of R1) as just discussed and the overlying Akraban Trachyandesite (near the base of N1) at 29.61 ± 0.08 Ma. We thus estimate the age of this reversal at 29.8 ± 0.2 Ma. The lower basaltic flows span 30.51 ± 0.36 to 29.94 ± 0.24 Ma, a statistically unresolvable interval of time contained within the reversed R1 zone.

6.5. Environmental impact of Afro-Arabian flood volcanism

Several papers have discussed the temporal coincidence between LIP volcanism and global environmental changes [2,3]. The Afro-Arabian LIP has been suggested to have a causal relationship with the Eocene–Oligocene biotic extinctions [58]. Extensive geochronologic and magnetostratigraphic studies in Ethiopia demonstrate that the Afro-Arabian LIP post-date the Eocene–Oligocene boundary, however, and a potential temporal link to the global Oi2 cooling event at about 30 Ma has instead been suggested [4,5]. According to the chronology of Pekar and Miller [59] the Oi2 cooling event takes place at the base of Chron C11r and therefore is synchronous with the onset of Afro-Arabian basaltic volcanism. Since the bulk of Afro-Arabian basaltic volcanism seems to have been rapidly erupted, a causal link to the Oi2 event cannot be ruled out. From other LIPs there is preponderance of evidence that basaltic LIP volcanism relates to global climate perturbations through volatile emissions [3]. The massive silicic Afro-Arabian volcanism is also a likely candidate to drive global cooling through massive airborne particulate ejection i.e., the "volcanic winter" mechanism as some of the ignimbrite units are found in Indian Ocean sediments, 2700 km away from the Afro-Arabian traps, implying that these units were emplaced on a near-global scale and that individual eruptive volumes may have been larger than 3000 km^3 dense rock equivalent [14]. However, these silicic units were erupted episodically over a period of ~3 Ma allowing potential climate recovery between eruptive episodes, and the main silicic Afro-Arabian volcanism occur in C11n and therefore post-date the Oi2 cooling event.

7. Conclusions

In this paper, we have documented that

(i) The paleomagnetic pole of this study $\lambda = 74.2°N$, $\varphi = 249.1°E$ ($A95 = 3.6°$; $N = 48$) is in excellent accord with the paleomagnetic pole obtained from the Ethiopian part of the Afro-Arabian province, after closure of the Red Sea.

(ii) Analysis of this new paleomagnetic pole in conjunction with contemporaneous paleomagnetic poles available from different latitudes suggest that Oligocene paleomagnetic field was dominated by the axial dipole with only insignificant non-dipole field contributions.

(iii) The massive rhyolitic ignimbrite eruptions, correlated to ash layers in Oligocene Indian Ocean sediment 2700 km away from the Afro-Arabian traps [56], took place during magnetochron C11n.

(iv) Silicic and bi-modal volcanism preserved on the Arabian Peninsula was episodic over ~3 Ma and seems to post-date the Oi2 global cooling event.

(v) The bulk of basal basaltic volcanism in Yemen took place in <1 Ma during a reverse polarity chron that appears to correspond to C11r and cannot be ruled out as having a causal role in the Oi2 global cooling event.

Acknowledgments

Funding for this research was provided by the Danish National Research Foundation, and the Carls-

berg Foundation. Tam Hagensen assisted the paleomagnetic field work. In our analysis we used software of Lisa Tauxe, Randy Enkin, Trond Torsvik and Mark Smethurst.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.epsl.2005.06.016.

References

[1] R.E. Ernst, K.L. Buchan, Recognizing mantle plumes in the geological record, Annu. Rev. Earth Planet. Sci. 31 (2003) 469–523.
[2] V. Courtillot, P.R. Renne, On the ages of flood basalt events, C.R. Geoscience 335 (2003) 113–140.
[3] P.B. Wignall, Large igneous provinces and mass extinctions, Earth-Sci. Rev. 53 (2001) 1–33.
[4] P. Rochette, E. Tamrat, G. Feraud, R. Pik, V. Courtillot, E. Ketofo, C. Coulon, C. Hofmann, D. Vandamme, G. Yirgu, Magnetostratigraphy and timing of the Oligocene Ethiopian traps, Earth Planet. Sci. Lett. 164 (1998) 497–510.
[5] C. Hofmann, V. Courtillot, G. Feraud, P. Rochette, G. Yirgu, E. Ketefo, R. Pik, Timing of the Ethiopian flood basalt event and implication for plume birth and global change, Nature 369 (1997) 838–841.
[6] I.A. Ukstins, P.R. Renne, E. Wolfenden, J. Baker, D. Ayalew, M. Menzies, Matching conjugate volcanic rifted margins: $^{40}Ar/^{39}Ar$ chrono-stratigraphy of pre- and syn-rift bimodal flood volcanism in Ethiopia and Yemen, Earth Planet. Sci. Lett. 198 (2002) 289–306.
[7] E. Coulie, X. Quidelleur, P.Y. Gillot, V. Courtillot, J.C. Lefevre, S. Chiesa, Comparative K–Ar and Ar/Ar dating of Ethiopian and Yemenite Oligocene volcanism: implications for timing and duration of the Ethiopian traps, Earth Planet. Sci. Lett. 206 (2003) 477–492.
[8] M.A. Richards, R.A. Duncan, V.E. Courtillot, Flood basalts and hot-spot tracks — plume heads and tails, Science 246 (1989) 103–107.
[9] J. Hopp, M. Trieloff, R. Altherr, Neon isotopes in mantle rocks from the Red Sea region reveal large-scale plume–lithosphere interaction, Earth Planet. Sci. Lett. 219 (2004) 61–76.
[10] V. Courtillot, C. Jaupart, I. Manighetti, P. Tapponnier, J. Besse, On causal links between flood basalts and continental breakup, Earth Planet. Sci. Lett. 166 (1999) 177–195.
[11] J. Ritsema, R.M. Allen, The elusive mantle plume, Earth Planet. Sci. Lett. 207 (2003) 1–12.
[12] P. Mohr, B. Zanettin, The Ethiopian flood basalt province, in: J.D. Macdougall (Ed.), Continental Flood Basalts, Kluwer, Dordrecht, 1988, pp. 63–110.
[13] J. Baker, L. Snee, M. Menzies, A brief Oligocene Period of flood volcanism in Yemen: implications for the duration and rate of continental flood volcanism at the Afro-Arabian triple junction, Earth Planet. Sci. Lett. 138 (1996) 39–55.
[14] I. Ukstins Peate, J.A. Baker, M. Al-Kadasi, A. Al-Subbary, K.B. Knight, P. Riisager, M.F. Thirlwall, D.W. Peate, P.R. Renne, M.A. Menzies, Volcanic stratigraphy of large-volume silicic pyroclastic eruptions during Oligocene Afro-Arabian flood volcanism in Yemen, Bull. Volcanology (in press).
[15] J.L. Kirchvink, The least-squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc. 62 (1980) 699–718.
[16] P.R. Renne, C.C. Swisher, A.L. Deino, D.B. Karner, T.L. Owens, D.J. DePaolo, Intercalibration of standards, absolute ages and uncertainties in $^{40}Ar/^{39}Ar$ dating, Chem. Geol. 145 (1998) 117–152.
[17] K.B. Knight, P.R. Renne, A. Halkett, N. White, $^{40}Ar/^{39}Ar$ dating of the Rajahmundry Traps, eastern India and their relationship to the Deccan Traps, Earth Planet. Sci. Lett. 208 (2003) 85–99.
[18] J. Riisager, P. Riisager, A.K. Pedersen, Paleomagnetism of large igneous provinces: case-study from west Greenland, North Atlantic igneous province, Earth Planet. Sci. Lett. 214 (2003) 409–426.
[19] K.B. Knight, S. Nomade, P.R. Renne, A. Marzoli, H. Bertrand, N. Youbi, The central Atlantic magmatic province at the Triassic–Jurassic boundary: paleomagnetic and $^{40}Ar/^{39}Ar$ evidence from Morocco for brief, episodic volcanism, Earth Planet. Sci. Lett. 228 (2004) 143–160.
[20] D. Vandamme, A new method to determine paleosecular variation, Phys. Earth Planet. Inter. 85 (1994) 131–142.
[21] P.L. McFadden, M.W. McElhinny, Classification of the reversal test in palaeomagnetism, Geophys. J. Int. 103 (1990) 725–729.
[22] P.L. McFadden, R.T. Merrill, M.W. McElhinny, S. Lee, Reversals of the Earth's magnetic field and temporal variations of the dynamo family, J. Geophys. Res. 96 (1991) 3923–3933.
[23] A. Cox, Confidence limits for the precision parameter k, Geophys. J. R. Astron. Soc. 18 (1969) 545–549.
[24] G.S. Watson, R.J. Enkin, The fold test in paleomagnetism as a parameter-estimation problem, Geophys. Res. Lett. 20 (1993) 2135–2137.
[25] M.F. Knudsen, B.H. Jacobsen, N. Abrahamsen, Palaeomagnetic distortion modelling and possible recovery by inversion, Phys. Earth Planet. Inter. 135 (2003) 55–73.
[26] P.R. Renne, R.C. Walter, K.L. Verosub, M. Sweitzer, J.L. Aronson, New data from Hadar (Ethiopia) support orbitally tuned time scale to 3.3 million years ago, Geophys. Res. Lett. 20 (1993) 1067–1070.
[27] T. Kidane, B. Abebe, V. Courtillot, E. Herrero, New paleomagnetic result from the Ethiopian flood basalts in the Abbay (Blue Nile) and Kessem gorges, Earth Planet. Sci. Lett. 203 (2002) 353–367.
[28] K.S. Kellogg, R.L. Reynolds, Opening of the Red Sea: constraints from a palaeomagnetic study of the As Sarat volcanic field, south-western Saudi Arabia, Geophys. J. R. Astron. Soc. 74 (1983) 649–665.

[29] A.Y. Izzeldin, Seismic, gravity and magnetic surveys in the central parts of the Red Sea, their interpretation and implications for the structure and evolution of the Red Sea, Tectonophysics 143 (1987) 269–306.
[30] D.P. McKenzie, D. Davies, P. Molnar, Plate tectonics of Red-Sea and east Africa, Nature 226 (1970) 243–248.
[31] R. Freund, Plate tectonics of the Red Sea and east Africa, Nature 228 (1970) 453.
[32] X. LePichon, J.M. Gaulier, The rotation of Arabia and Levant fault system, Tectonophysics 153 (1988) 271–294.
[33] S. Joffe, Z. Garfunkel, Plate kinematics of the circum Red Sea — a re-evaluation, Tectonophysics 141 (1987) 5–22.
[34] M. Sultan, R. Becker, R.E. Arvidson, P. Shore, R.J. Stern, Z. El Alfy, R.I. Attia, New constraints on Red Sea rifting from correlations of Arabian and Nubian Neo-Proterozoic outcrops, Tectonics 12 (1993) 1303–1319.
[35] B. Collet, H. Taud, J.F. Parrot, F. Bonavia, J. Chorowicz, A new kinematic approach for the Danakil block using a digital elevation model representation, Tectonophysics 316 (2000) 343–357.
[36] T.H. Torsvik, R. Van der Voo, J.G. Meert, J. Mosar, H.J. Walderhaug, Reconstructions of the continents around the North Atlantic at about the 60th parallel, Earth Planet. Sci. Lett. 187 (2001) 55–69.
[37] M.E. Beck, B.A. Housen, Pre-drift extension of the Atlantic margins of North America and Europe based on paths of Permo-Triassic apparent polar wander, Geophys. J. Int. 152 (2003) 68–78.
[38] J. Carlut, V. Courtillot, How complex is the time-averaged geomagnetic field over the past 5 Myr? Geophys. J. Int. 134 (1998) 527–544.
[39] R.T. Merril, P.L. McFadden, The geomagnetic axial dipole field assumption, Phys. Earth Planet. Inter. 139 (2003) 171–185.
[40] D.V. Kent, M.A. Smethurst, Shallow bias of paleomagnetic inclinations in the Paleozoic and Precambrian, Earth Planet. Sci. Lett. 160 (1998) 391–402.
[41] J. Besse, V. Courtillot, Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr, J. Geophys. Res. 107 (2002) 2300, doi:10.1029/2000JB000050.
[42] J.A. Tarduno, R.D. Cottrell, A.V. Smirnov, The Cretaceous superchron geodynamo: observations near the tangent cylinder, Proc. Natl. Acad. Sci. U. S. A. 99 (2002) 14020–14025.
[43] D.A. Schneider, D.V. Kent, Testing models of the Tertiary paleomagnetic field, Earth Planet. Sci. Lett. 101 (1990) 260–271.
[44] J.W. Si, R. Van der Voo, Too-low magnetic inclinations in central Asia: an indication of a long-term Tertiary non-dipole field? Terra Nova 13 (2001) 471–478.
[45] W. Krijgsman, L. Tauxe, Shallow bias in Mediterranean paleomagnetic directions caused by inclination error, Earth Planet. Sci. Lett. 222 (2004) 685–695.
[46] S. Gilder, Y. Chen, J.P. Cogne, X.D. Tan, V. Courtillot, D.J. Sun, Y.G. Li, Paleomagnetism of Upper Jurassic to Lower Cretaceous volcanic and sedimentary rocks from the western Tarim Basin and implications for inclination shallowing and absolute dating of the M-0 (ISEA?) chron, Earth Planet. Sci. Lett. 206 (2003) 587–600.
[47] R.F. Butler, G.E. Gehrels, S.L. Baldwin, C. Davidson, Paleomagnetism and geochronology of the Ecstall pluton in the Coast Mountains of British Columbia: evidence for local deformation rather than large-scale transport, J. Geophys. Res. 107 (2002) 2009, doi:10.1029/2001JB000270.
[48] S.C. Cande, D.V. Kent, Revised calibration of the geomagnetic polarity timescale for the Late Cretaceous and Cenozoic, J. Geophys. Res. 100 (1995) 6093–6095.
[49] S.P. Huestis, G.D. Acton, On the construction of geomagnetic timescales from non-prejudicial treatment of magnetic anomaly data from multiple ridges, Geophys. J. Int. 129 (1997) 176–182.
[50] W.A. Berggren, D.V. Kent, J.J. Flyn, J.A. van Couvering, Cenozoic geochronology, Geol. Soc. Amer. Bull. 96 (1985) 1407–1418.
[51] N.J. Shackleton, M.A. Hall, I. Raffi, L. Tauxe, J. Zachos, Astronomical calibration age for the Oligocene–Miocene boundary, Geology 28 (2000) 447–450.
[52] G.S. Odin, A. Montanari, A. Deino, R. Drake, P.G. Guise, H. Kreuzer, D.C. Rex, Reliability of volcano-sedimentary biotite ages across the Eocene–Oligocene boundary (Apennines, Italy), Chem. Geol. 86 (1991) 203–224.
[53] F. Begemann, K.R. Ludwig, G.W. Lugmair, K. Min, L.E. Nyquist, P.J. Patchett, P.R. Renne, C.-Y. Shih, I.M. Villa, R.J. Walker, Call for an improved set of decay constants for geochronological use, Geochim. Cosmochim. Acta 65 (2001) 111–121.
[54] W.A. Berggren, D.R. Prothero, Eocene–Oligocene climatic and biotic evolution: an overview, in: D.R. Prothero, W.A. Berggren (Eds.), Eocene–Oligocene Climatic and Biotic Evolution, Princeton Series in Geology and Paleonotology, 1992, pp. 1–28.
[55] W.A. Berggren, D.V. Kent, C.C. Swisher III, M.-P. Aubry, A revised Cenozoic geochronology and chronostratigraphy, SEPM Spec. Pub. 54 (1995) 129–212.
[56] I. Ukstins Peate, J.A. Baker, A.J.R. Kent, M. Al-Kadasi, A. Al-Subbary, D. Ayalew, M. Menzies, Correlation of Indian Ocean tephra to individual Oligocene silicic eruptions from Afro-Arabian flood volcanism, Earth Planet. Sci. Lett. 211 (2003) 311–327.
[57] Y. Touchard, P. Rochette, M.P. Aubry, A. Michard, High-resolution magnetostratigraphic and biostratigraphic study of Ethiopian traps-related products in Oligocene sediments from the Indian Ocean, Earth Planet. Sci. Lett. 206 (2003) 493–508.
[58] V. Courtillot, Mass extinctions in the last 300 million years: one impact and seven flood basalts? Isr. J. Earth Sci. 43 (1994) 255–266.
[59] S. Pekar, K.G. Miller, New Jersey Oligocene "Icehouse" sequences (ODP Leg 150X) correlated with Global d18O and Exxon eustatic records, Geology 24 (1996) 567–570.
[60] R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. Lond., A 217 (1953) 295–305.
[61] R.H. Steiger, E. Jager, Subcommission on geochronology — convention on use of decay constants in geochronology and cosmochronology, Earth Planet. Sci. Lett. 36 (1977) 359–362.

[62] K.W. Min, R. Mundil, P.R. Renne, K.R. Ludwig, A test for systematic errors in Ar-40/Ar-39 geochronology through comparison with U/Pb analysis of a 1.1-Ga rhyolite, Geochim. Cosmochim. Acta 64 (2000) 73–98.
[63] G. Audi, O. Bersillon, J. Blachot, A.H. Wapstra, The NUBASE evaluation of nuclear and decay properties, Nucl. Phys., A 624 (1997) 1–124.
[64] P.R. Renne, E.B. Norman, Determination of the half-life of Ar-37 by mass spectrometry, Phys. Rev., C Nucl. Phys. 63 (2001) (Art. No. 047302).
[65] R.D. Muller, W.R. Roest, J.Y. Royer, L.M. Gahagan, J.G. Sclater, Digital isochrons of the world's ocean floor, J. Geophys. Res. 102 (1997) 3211–3214.
[66] G. Plenier, P. Camps, B. Henry, K. Nicolaysen, Palaeomagnetic study of Oligocene (24–30 Ma) lava flows from the Kerguelen Archipelago (southern Indian Ocean): directional analysis and magnetostratigraphy, Phys. Earth Planet. Inter. 133 (2002) 127–146.
[67] A.G. Schmidt, P. Riisager, N. Abrahamsen, J. Riisager, A.K. Pedersen, R. van der Voo, Paleomagnetism of Eocene Talerua Member lavas on Hareøen Island, West Greenland, Bull. Geol. Soc. Den. 52 (2005).
[68] M.W. McElhinny, B.J. Embleton, P. Wellman, Synthesis of Australian Cenozoic paleomagnetic results, Geophys. J. R. Astron. Soc. 36 (1974) 141–151.
[69] N. Weinreich, U. Bleil, A paleomagnetic study of Tertiary volcanics from the Hocheifel (Germany) and the Cenozoic apparent polar wander path of Central-Europe, J. Geophys. 54 (1984) 163–173.